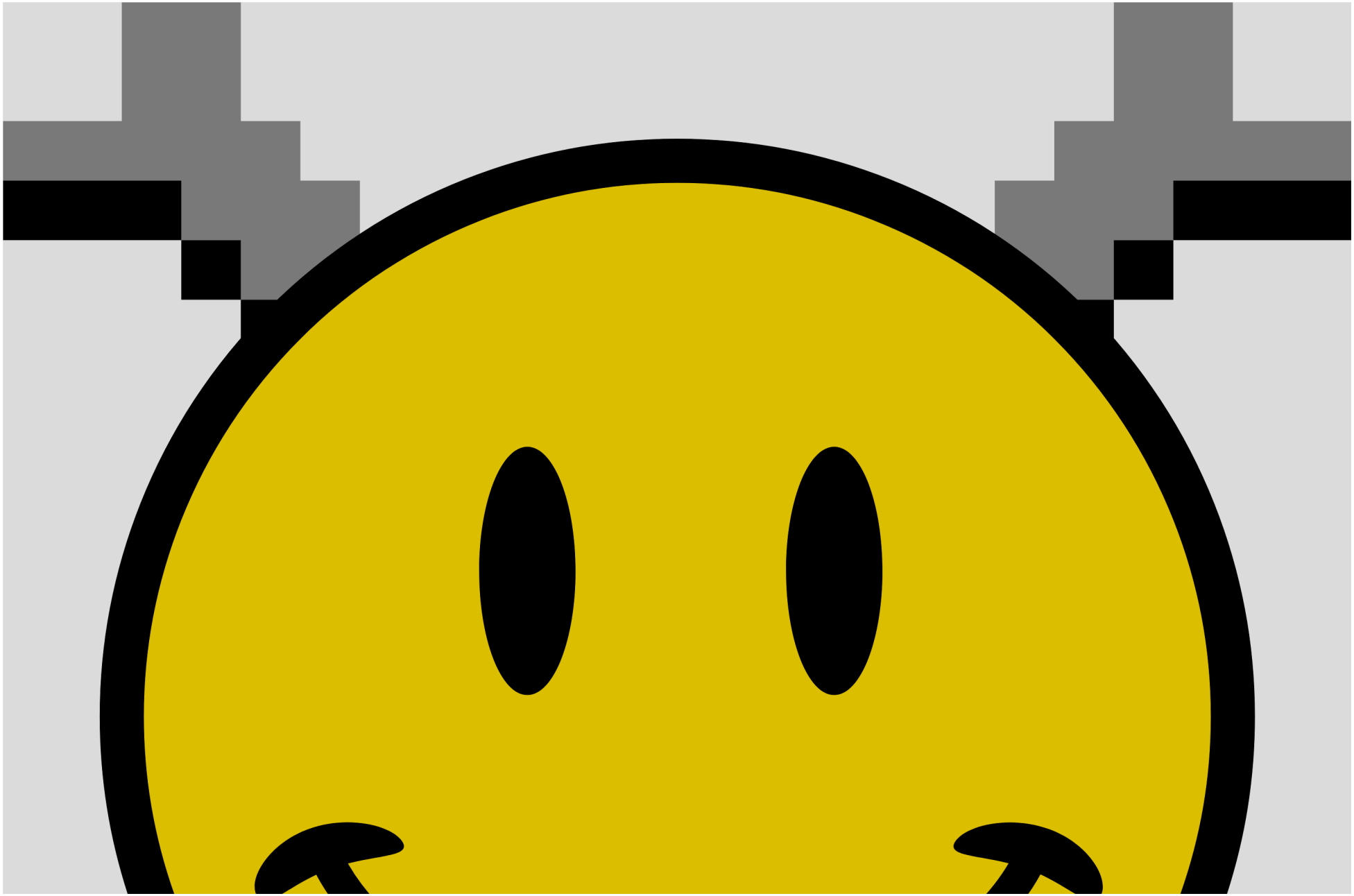
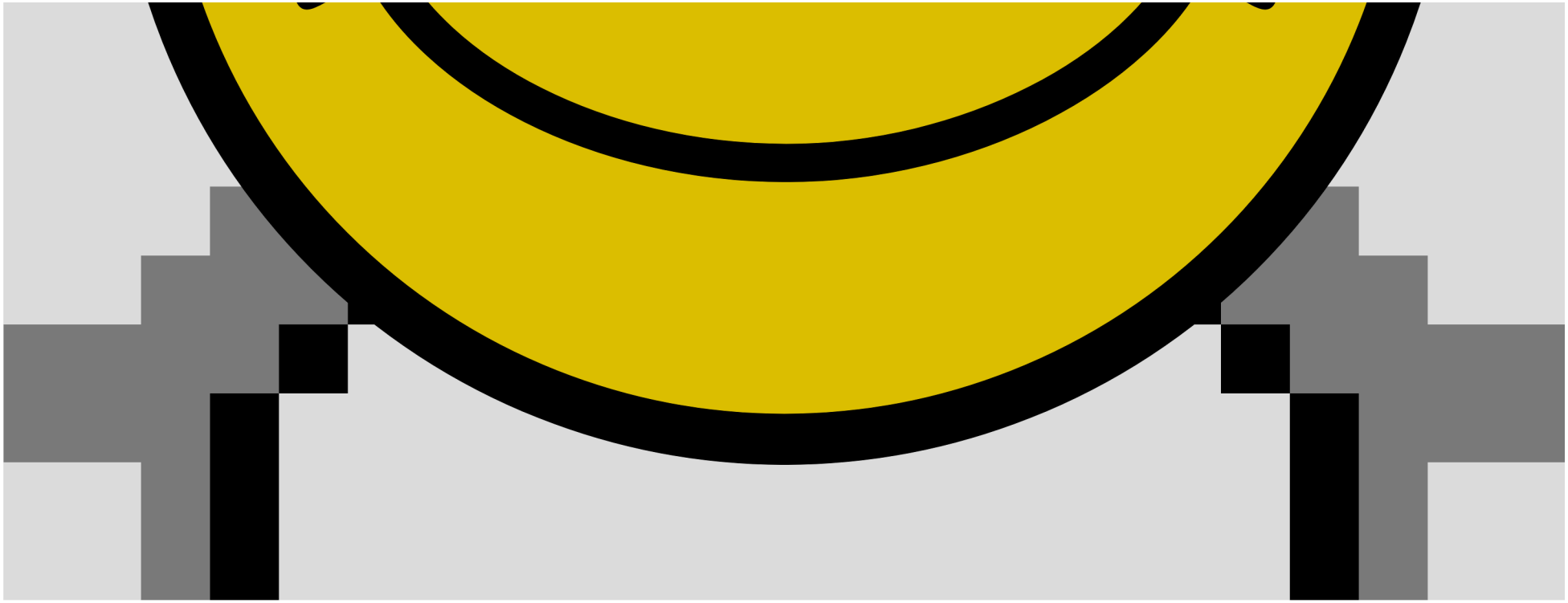




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AI Infrastructure Sovereignty

Sergio Cruzes

Optical Network Engineering, Ciena Brazil, Ciena, Av. das Nações Unidas, 14.171 – 15º andar – Marble Tower – Salas 1563/1564, São Paulo, 04794-000, SP, Brazil The views, analyses, and conclusions expressed in this paper are those of the author alone and do not represent the position, policy, or endorsement of Ciena Corporation or any of its affiliates. This research was conducted by the author independently in a personal academic capacity.

Abstract

Artificial intelligence is no longer just a software problem. It has become an infrastructure problem. Training and inference at scale now depend on tightly connected data centers, high-capacity optical networks, and energy systems operating close to their physical and environmental limits. In this context, control over data and algorithms is not enough. Real AI sovereignty depends on the ability to deploy, operate, and adapt infrastructure under constraints such as energy availability, sustainability requirements, and network reach.

This tutorial-survey introduces the concept of AI infrastructure sovereignty, defined as the ability of a region, operator, or nation to maintain operational control over AI systems within these constraints. The central idea is that sovereignty emerges from the joint design of three layers: AI-oriented data centers, optical transport networks, and control frameworks that provide real-time visibility and coordination across them.

We first examine how AI workloads are reshaping data center design, pushing power densities higher, increasing cooling demands, and tightening the relationship with local energy systems. In this setting, factors such as carbon intensity and water usage become hard limits on where and how AI can be deployed. We then look at optical networks as the backbone of distributed AI, showing how latency, capacity, failure domains, and jurisdictional boundaries directly influence what can be achieved in practice.

Building on this foundation, the paper highlights the role of telemetry, agentic AI, and digital twins as key enablers of operational sovereignty. Together, they make it possible to monitor, coordinate, and validate system behavior across compute, network, and energy domains in a closed loop.

The tutorial concludes with a reference architecture for sovereign AI infrastructure that integrates telemetry pipelines, agent-based control, and digital twins, treating sustainability as a core design constraint rather than an afterthought.

Keywords: : AI infrastructure sovereignty, data center, sustainability, telemetry, agentic AI

1 Introduction

Artificial intelligence has become a defining force in the evolution of digital infrastructure. What once progressed mainly through advances in algorithms and software is now increasingly shaped by physical constraints: power availability, cooling capacity, water usage, optical connectivity, and integration with the electrical grid. Large-scale AI systems already operate at the scale of industrial facilities, with data centers drawing tens to hundreds of megawatts per site and exhibiting rapid, high-amplitude power fluctuations that challenge both facility design and grid stability [53, 38, 12]. As a result, the limits on AI progress are no longer primarily computational, but infrastructural.

Modern AI workloads, especially large language models, rely on tightly coordinated computation across dense clusters of accelerators, supported by high-capacity data centers and low-latency optical networks. Training concentrates extreme power and cooling demand over sustained periods, while inference runs continuously at global scale, often dominating the overall energy and water footprint [33, 62]. These characteristics tightly couple com-

pute, network, and energy systems, making performance, resilience, and sustainability inseparable from how infrastructure is designed and operated.

1.1 From software sovereignty to infrastructure sovereignty

These realities call for a shift in how AI sovereignty is understood. Most existing discussions focus on data localization, jurisdiction, privacy, and model ownership. Frameworks such as GDPR [26] and the CLOUD Act [72] show how sovereignty has been approached through legal and regulatory mechanisms. Intellectual property protections over data, models, and software add another layer of formal control. While these remain important, they do not fully explain who can actually deploy, operate, and evolve AI systems at scale [29, 28].

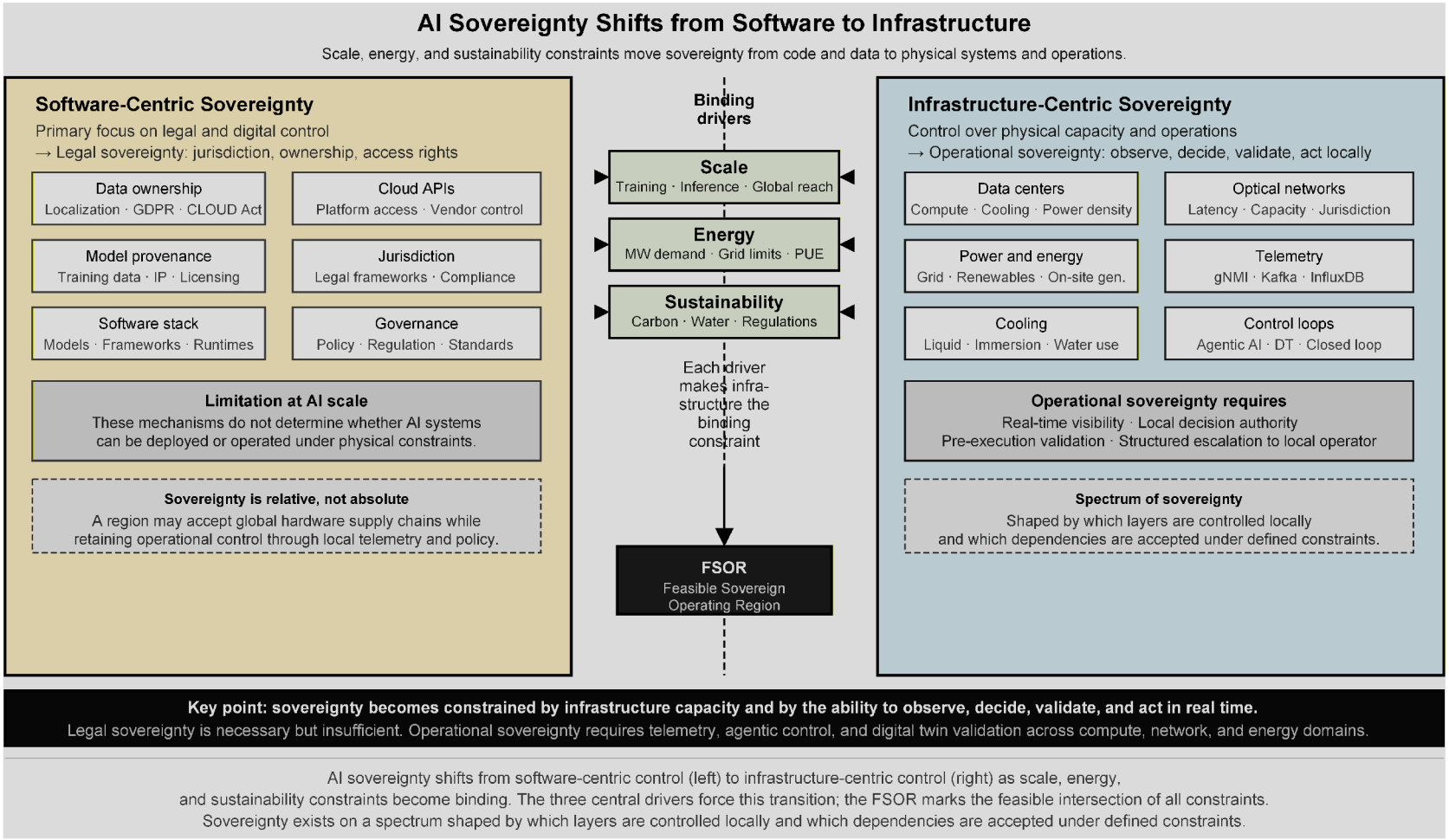
In practice, sovereignty increasingly depends on access to physical resources such as electricity, water, land, fiber infrastructure, and automation capabilities, along with the ability to coordinate them under real operational constraints.

This tutorial draws a key distinction between *legal sovereignty* and *operational sovereignty*. Legal sovereignty concerns jurisdiction, compliance, and formal ownership, as reflected in regulatory frameworks and IP protections. Operational sovereignty, by contrast, is about what can actually be done: the ability to observe system state, make decisions based on local conditions, validate those decisions, and act within defined policies and physical limits. It depends on visibility, control authority, and the ability to execute reliably across compute, network, and energy sys-

tems. In this sense, telemetry, automation, and closed-loop control are not just operational tools, but core enablers of sovereign AI infrastructure.

Sustainability adds another dimension to this picture. It is not just a reporting requirement, but a real constraint that must be enforced in operation. Metrics such as Power Usage Effectiveness (PUE) [68], which measures the overhead of cooling and power delivery relative to IT load, become critical. As rack power densities increase, even small inefficiencies translate into large absolute energy waste, with direct implications for carbon emissions and water consumption.

Figure 1 captures this transition. Traditional views of sovereignty are centered on software and legal control: data ownership, model provenance, intellectual property (IP), and regulatory compliance. In this context, IP refers to the legal ownership of training data, model weights, algorithms, and software stacks, including the rights to use, license, and restrict access to them. These mechanisms define who formally controls AI assets, but they do not by themselves determine whether AI systems can be deployed or operated at scale. An infrastructure-centric perspective shifts the focus to physical and operational constraints, including power capacity, cooling efficiency, optical connectivity, and real-time system visibility. Three factors drive this transition: scale, energy, and sustainability. Each imposes limits on where and how AI systems can operate, and together they define what this tutorial calls the Feasible Sovereign Operating Region (FSOR), introduced in Section 7. Sovereignty, in this sense, is not absolute. It exists on a spectrum shaped by which parts of the stack are controlled locally and which dependencies are accepted under clearly defined constraints [28].



AI sovereignty shifts from software-centric control (left) to infrastructure-centric control (right) as scale, energy, and sustainability constraints become binding. The three central drivers force this transition; the FSOR marks the feasible intersection of all constraints. Sovereignty exists on a spectrum shaped by which layers are controlled locally and which dependencies are accepted under defined constraints.

Figure 1: **AI sovereignty shifts from software-centric to infrastructure-centric control as scale, energy, and sustainability constraints become binding.** The left panel shows software-centric sovereignty — legal control over data (GDPR, CLOUD Act), model IP, jurisdiction, and governance — which is necessary but insufficient at AI scale. The three central drivers (scale at MW-class power demand, energy including grid limits and PUE, and sustainability including carbon, water, and regulations) force a transition: each makes physical infrastructure the binding constraint

on AI deployment and operation, independently of software capability or regulatory compliance. Their intersection defines the Feasible Sovereign Operating Region (FSOR) — the space within which all three constraints can be jointly satisfied. The right panel shows infrastructure-centric sovereignty — operational control over data centers, optical networks, power and energy systems, telemetry, cooling, and closed-loop control systems (agentic AI, digital twins, and closed-loop automation) — which is the practical expression of operational sovereignty. Sovereignty exists on a spectrum shaped by which layers are controlled locally and which dependencies are accepted under defined constraints [28, 44].

1.2 Infrastructure as the binding constraint

Recent geopolitical developments reinforce this view. Export controls, sanctions, and concentration of platforms have shown that access to compute and infrastructure can be restricted by external decisions [24]. As AI becomes central to economic activity and national security, many regions are seeking ways to reduce this exposure and achieve greater autonomy [44].

The physical demands of AI workloads make infrastructure the key constraint. Training clusters can require hundreds of megawatts, generate extreme heat densities, and introduce rapid power fluctuations that challenge both facilities and grid interfaces [53, 12]. Inference workloads run continuously at large scale, increasing total energy, water, and network demand over time [33, 52]. In this context, data centers become strategic assets, and optical networks become critical control surfaces that define latency, reach, and resilience [42, 62].

Regions with limited grid capacity, high carbon intensity, water constraints, or insufficient connectivity face real barriers to deploying AI systems, regardless of their software capabilities [38, 12, 74]. On the other hand, regions that can align clean energy, efficient cooling, and high-capacity optical transport gain a lasting advantage. This makes it clear that models of sovereignty based only on data or software assume an infrastructure that is effectively unlimited and interchangeable. That assumption no longer holds. Compute cannot be moved freely without considering power, cooling, latency, and reliability constraints [42, 21].

AI infrastructure sovereignty can therefore be understood as the ability to design, operate, and control AI systems within these constraints, without excessive reliance on external actors [34]. In practical terms, it comes down to operational control: being able to observe the system, make decisions, validate them, and act within local limits and policies.

1.3 Telemetry and agentic AI as the operational layer

Achieving this level of control requires mechanisms that connect high-level intent to real infrastructure behavior. Telemetry provides continuous visibility into the state of data centers, networks, and energy systems [20, 7]. Agentic AI builds on this by making decisions, coordinating actions across domains, and enforcing policies under uncertainty [20, 40]. Digital twins provide a way to simulate and validate actions before they are applied, reducing risk in complex systems [20]. Large language models help translate human intent into structured objectives and explain system behavior in accessible terms [19].

Together, these elements form a closed-loop control system. Telemetry enables observation, agentic systems enable decision-making, digital twins provide validation, and execution systems carry out actions within defined constraints. This approach extends earlier work in network and data center automation to a broader setting that includes energy systems and sustainability constraints, forming a unified control framework for sovereign AI infrastructure.

1.4 Scope and organization

This tutorial provides a cross-layer view of AI infrastructure sovereignty grounded in engineering practice. It is intended as a system-level synthesis for researchers, operators, and policymakers. It brings together two areas that have often been treated separately: discussions of AI sovereignty focused on data and regulation, and work on AI infrastructure focused on scaling and efficiency. The central argument is that sovereignty in the AI era depends on the ability to observe, control, and optimize physical infrastructure under real constraints.

The rest of the paper is organized as follows. Section 2 shows how AI workloads translate into physical infrastructure constraints. Section 3 examines data centers and the sustainability limits that shape deployment. Section 4 analyzes optical networks and their role in defining latency, capacity, and control boundaries. Sections 5 and 6 introduce telemetry, agentic AI, and digital twins as the operational foundation. Section 7 presents a reference architecture, and Section 8 discusses sustainability-aware automation and its policy implications.

2 From AI Models to Physical Infrastructure

AI infrastructure cannot be understood as a purely digital stack. At scale, it is shaped by a hierarchy of physical constraints that start with power availability and extend upward through cooling, networking, and operations. Higher layers can improve efficiency and utilization, but they cannot bypass limits imposed by electrical capacity, heat dissipation, or physical space. As AI systems grow, infrastructure design becomes less about abstraction and more about managing these constraints directly. This shift is central to sovereignty: control over models and data is necessary, but it does not matter if the underlying systems cannot be built, powered, or cooled.

2.1 The infrastructure demands of modern AI workloads

AI progress is often described in terms of model size, data, and algorithmic advances. But behind that narrative is a deeper change in how workloads behave. Modern AI systems, especially large-scale foundation models, rely on tightly synchronized computation across thousands of accelerators acting as a single system [9, 53]. Training concentrates extreme demand into short, intense periods, while inference spreads continuous demand across global platforms [33, 62]. Together, these patterns make AI far more dependent on physical infrastructure than traditional software workloads [4].

Figure 2 illustrates how this transition happens. Software-level characteristics, such as synchronization, latency sensitivity, and bandwidth requirements, propagate downward into concrete infrastructure demands. What begins as a property of the workload becomes a requirement for power delivery, cooling capacity, space, and network bandwidth. In other words, workload behavior turns into physical constraint.

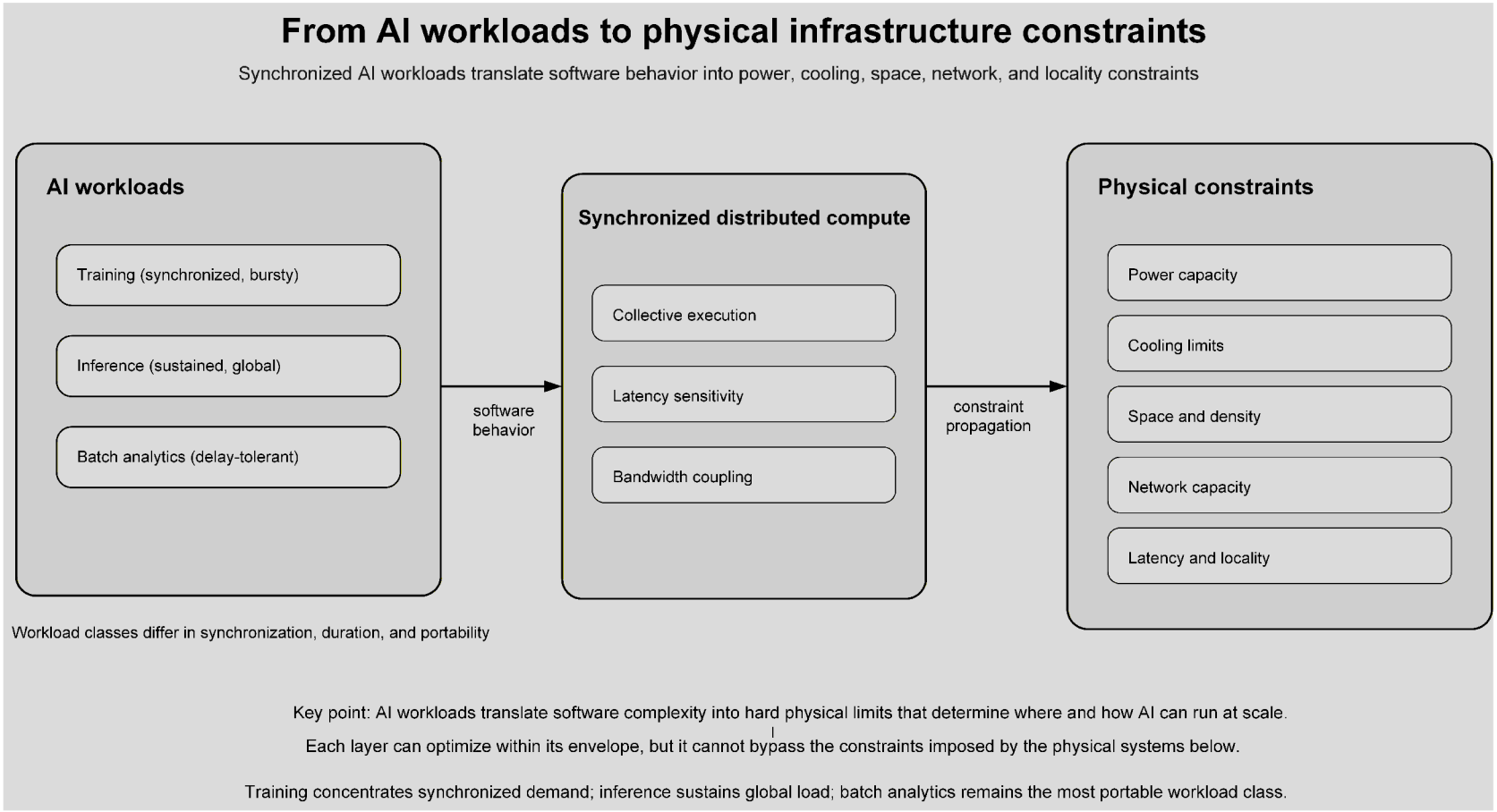


Figure 2: From AI workloads to physical infrastructure constraints. Synchronized training and large-scale inference translate software demands into hard limits on power delivery, cooling capacity, physical space, and network bandwidth. Each layer can optimize within its envelope but cannot bypass the constraints imposed by the layer below.

2.2 Power, cooling, and space as primary constraints

Once compute is tightly synchronized, pressure builds across all supporting systems at the same time. Power demand is not only high, but highly dynamic. Rapid fluctuations driven by collective communication patterns challenge conventional power distribution and backup systems [14, 12]. Cooling must handle extreme heat densities, often beyond what air-based systems can manage, which is why liquid cooling and immersion solutions are becoming standard [62, 38]. Physical space adds another constraint: dense racks, power equipment, and cooling systems compete for limited floor area.

At this point, the limiting factor is no longer just the availability of accelerators, but whether a site can support them. Grid capacity, transformers, cooling infrastructure, and water availability all place hard limits on how much AI capacity can be deployed at a given location [38, 12]. These limits vary widely across regions, creating uneven conditions for AI deployment that software alone cannot overcome.

This marks a shift from earlier cloud models, where workloads could be moved relatively freely between data centers. AI workloads are far less portable. They tie computation to locations that can meet strict power, cooling, and networking requirements, exposing operators directly to local infrastructure constraints [62]. Regions with reliable energy, efficient cooling, and strong optical connectivity can support large-scale AI systems and expand over time [38]. Others face structural barriers that cannot be solved through software optimization. Even when hardware is available, a lack of power or cooling makes it unusable.

2.3 Workload classes and their infrastructure signatures

Not all AI workloads place the same demands on infrastructure. Table 1 compares three main classes: training, inference, and batch analytics. This distinction matters for both sovereignty and sustainability planning. Training workloads are tightly coupled and must run within strict latency bounds, making them hard to relocate. Inference workloads are more distributed, but still need to remain close to users to meet latency requirements. Batch analytics workloads are more flexible and can be shifted in time or location, making them the main candidates for adapting to energy or sustainability constraints.

Table 1: AI workload classes and their infrastructure signatures. Batch analytics workloads are delay-tolerant and the primary candidates for sustainability-driven temporal and spatial shifting. Training workloads are spatially constrained by synchronization requirements and cannot be shifted without violating collective communication latency bounds.

AI workload	Power profile	Cooling stress	Network demand	Portability
Training	Bursty, high	Extreme	Very high	Low
Inference	Sustained	Moderate–high	High	Medium
Batch analytics	Variable, schedulable	Moderate	Low–moderate	High

Entries are qualitative relative comparisons within each column rather than absolute measurements; absolute values depend on cluster size, hardware generation, and deployment configuration.

In practical terms, portability refers to how easily a workload can be moved without breaking its performance constraints. Training has very low portability because of strict synchronization requirements. Inference has moderate portability, since it can be replicated but must stay close to demand. Batch workloads are highly portable and can be shifted to optimize for energy availability or carbon intensity. From a sovereignty perspective, this matters: only portable workloads can be moved to adapt to constraints. Non-portable workloads require local infrastructure to already be in place.

2.4 Infrastructure as a sovereignty boundary

This hierarchy of physical constraints has a direct impact on sovereignty. Control over AI systems is not determined only by data or models, but by who can provide power, cooling, connectivity, and operational control. Regions without sufficient grid capacity, water, or fiber infrastructure face limits that are physical rather than legal [71, 38].

Sovereignty therefore emerges unevenly across infrastructure layers. The workload distinctions in Table 1 make this clear. A region might support batch workloads but not large-scale training, meaning it has only partial AI sovereignty. Closing this gap requires investment not in software, but in energy systems, cooling, and optical networks. The next sections examine these constraint domains in more detail, starting with data center sustainability limits.

3 AI-Oriented Data Centers and Sustainability Limits

One of the central challenges in AI infrastructure planning is the mismatch between how quickly digital systems can scale and how slowly physical infrastructure evolves. Compute capacity can be deployed relatively fast, but power grids, substations, cooling systems, and permitting processes move on much longer timelines. Because of this mismatch, capital alone does not determine where AI can be built. Physical infrastructure becomes the limiting factor, and sustainability constraints, such as energy availability, carbon intensity, and water usage, are often where those limits show up most clearly.

3.1 How AI workloads reshape data center design

AI-oriented data centers are fundamentally different from traditional enterprise or cloud facilities. Large training and inference clusters pack a high concentration of accelerators into a single space, pushing rack power densities far beyond historical norms. This shift forces a move toward high-density layouts and requires a complete rethink of mechanical and electrical systems [62, 10].

Traditional data centers were designed for moderate and relatively stable heat loads. AI systems are not. They generate sustained, high-intensity heat and rapid load fluctuations driven by tightly synchronized computation. As a result, air-based cooling, while still used in hybrid setups, quickly reaches its limits. Once rack densities move beyond roughly 20–30 kW, which is now common in GPU clusters, air cooling alone cannot keep up without introducing impractical airflow requirements. This is why liquid cooling, both direct-to-chip and immersion, is becoming

standard. These approaches remove heat closer to the source and provide more stable thermal conditions under continuous AI workloads [38, 62].

What makes this shift particularly important is how tightly different systems become coupled. Power, cooling, and water usage are no longer independent design choices. Increasing electrical density raises heat, which determines the cooling approach, which in turn affects water consumption. A decision driven by compute demand can end up being limited by a water permit. Once air cooling is no longer viable, design moves from optimizing individual components to balancing trade-offs across the entire system. This is why sustainability constraints need to be considered together, not in isolation.

Figure captures this relationship. It shows how energy availability, carbon limits, and water and cooling feasibility jointly define the space in which AI deployment is possible. The feasible region, where all three constraints are satisfied, is not fixed. Over time, it is expected to shrink as water availability becomes more constrained [39], carbon regulations tighten [74, 38], and grid expansion struggles to keep pace with demand [38]. The boundaries shown in the figure are directional rather than precise, since the rate of change varies by region, grid mix, and local conditions. The implication is straightforward: a site that is viable today may not remain viable in the future without investment in clean energy, efficient cooling, or local generation.

3.2 Energy: the dominant constraint

Among all constraints, energy is the most immediate and often the most limiting. AI data centers require large, continuous power supplies, but they also need to handle rapid fluctuations caused by synchronized workloads [14, 12].

Grid interconnection capacity, transformer limits, and transmission constraints often determine how large a facility can be, even before any hardware is installed [62]. In many regions, long permitting cycles and slow grid upgrades further restrict how quickly capacity can expand.

There is also an important dynamic aspect. AI workloads do not draw power evenly. Collective operations during training can create sharp, short-lived spikes that exceed average consumption by a wide margin. This means that infrastructure must be sized for peak demand, not average usage. As a result, part of the available grid capacity cannot be fully utilized, increasing costs and reducing efficiency. Facilities that underestimate this behavior risk instability or failure during sustained workloads.

3.3 Carbon intensity: from preference to constraint

Carbon intensity has moved from being a sustainability preference to a real constraint on where AI infrastructure can be deployed. Operators are no longer concerned only with how much energy is available, but also with how that energy is produced. Regions with low-carbon grids or access to renewable generation are increasingly favored [17, 16]. Studies show that identical systems can have very different environmental impacts depending solely on the local energy mix [74].

In regions where energy comes primarily from fossil sources, AI expansion faces growing regulatory and financial pressure. Carbon reporting requirements and, in some cases, operational limits are becoming more common [38]. This turns carbon intensity into a site-level filter. Even if power and hardware are available, a site may not be eligible to host certain workloads if its emissions exceed defined thresholds. From a sovereignty perspective, this cre-

ates a new kind of barrier that cannot be solved with compute investment alone. It requires changes in energy sourcing or access to clean generation.

3.4 Water: the underestimated constraint

Water is often overlooked in AI infrastructure planning, but it can be just as limiting as energy. High-density cooling systems, especially those using evaporative methods, can consume large volumes of water per unit of compute [75]. Local climate plays a major role here. Temperature, humidity, and seasonal variation all affect cooling efficiency and water demand [38].

This introduces a strong seasonal effect. A facility that operates comfortably within its limits in winter may exceed its water permit during summer, when both cooling demand and water stress increase. In regions already facing water scarcity, this can make large AI deployments impractical, even when sufficient electrical capacity is available [74]. Unlike carbon, which can sometimes be managed through scheduling, water limits are typically fixed at the facility level and are harder to work around. Climate projections suggest that these constraints will become more severe over time, with longer and more intense dry periods [39].

3.5 Sustainability as a sovereignty boundary

Taken together, energy, carbon, and water define hard boundaries for AI deployment [38, 74]. These limits cannot be fully offset through incremental improvements in hardware or software. Once they are reached, further expan-

sion requires changes in infrastructure, not optimization.

This has direct implications for sovereignty. Control over AI is not just about data, models, or regulation. It depends on whether a region can provide the energy, cooling, and infrastructure needed to run these systems. A deployment may satisfy all legal requirements and still be infeasible because of physical constraints. In those cases, sovereignty is limited by infrastructure, not policy.

At the same time, regions that can combine reliable energy, efficient cooling, and strong connectivity gain a durable advantage. They can deploy and operate AI systems at scale and adapt over time. This leads to an uneven global landscape, where some regions can expand continuously while others face hard limits.

Table 2 summarizes these constraints, showing how each one operates, how it limits deployment, and what mitigation strategies are available. Together, they reinforce a central point: sustainability is not an optimization goal, but a feasibility condition.

Table 2: Sustainability constraints as deployment boundaries for AI-oriented data centers. Each constraint operates on a distinct timescale and requires different mitigation strategies.

Constraint	Metric	Mechanism	Timescale	Mitigation
Energy	MW, grid access	Limits scale; grid delays precede deployment	Years (grid)	On-site generation; demand response
Carbon	gCO_2eq/kWh	Determines site eligibility; regulatory exposure	Sub-hourly to annual	Renewable PPA; carbon-aware scheduling
Water	L/MWh	Limits cooling feasibility; seasonal constraints	Seasonal to decadal	Liquid cooling; water recycling

Ultimately, understanding these constraints is essential for assessing sovereign AI capacity. What matters is not just what can be designed in software, but what can actually be built and sustained in the physical world. The reach of these systems, and therefore the practical scope of sovereignty, is then determined by the optical networks that connect them, which are examined in the next section.

4 Optical Networks as the Backbone of AI Sovereignty

Optical networks are what tie AI data centers together into a functional system. While compute and storage stay anchored to specific sites, optical transport determines how those resources are connected, coordinated, and scaled across distance. In practice, optical infrastructure defines a sovereignty boundary of its own, separate from compute and energy.

Three factors explain why. First, the physics of propagation sets a hard lower bound on latency that cannot be reduced by better hardware or smarter routing [65]. Second, network capacity is limited by spectrum, transceivers, and power, which ultimately caps how much AI traffic can move between sites. Third, the physical paths that data follows determine jurisdictional exposure: which legal systems apply, who can monitor traffic, and who controls recovery when failures occur. Taken together, these factors mean that sovereignty depends not only on where data centers are located, but on how they are connected and governed [42].

4.1 Latency, capacity, and the physics of optical sovereignty

The speed of light in fiber, about 2×10^8 m/s, sets a propagation delay of roughly 5 ms per 1,000 km [65]. This is a hard limit. It does not improve with faster processors or better protocols. For AI workloads that depend on tight synchronization across accelerators, this latency directly defines how far apart those systems can be.

For example, if a training system can tolerate only 1 ms of communication delay, then participating nodes must be located within roughly 100 km of each other. Beyond that distance, the delay alone breaks synchronization, regard-

less of available bandwidth.

Capacity introduces another constraint. As AI systems scale, network requirements grow from petabit to exabit levels [61, 13]. At that point, optical capacity, transceiver density, and energy efficiency become limiting factors. If the network cannot move enough data, compute resources sit idle, no matter how powerful they are [69]. Limits in spectrum, port density, and power consumption all contribute to these ceilings.

Failures add a third dimension. Fiber cuts, equipment faults, and regional outages can isolate entire parts of the system [32]. How far those disruptions spread, and how quickly recovery happens, depends on the structure of the network. A system may meet every formal requirement for sovereignty and still be fragile if it depends on a small number of critical links or externally controlled routes [42, 20].

4.2 The scale-up and scale-out boundary

A key distinction in AI infrastructure is between scale-up and scale-out networks [13, 36]. Scale-out networks connect distributed systems and can tolerate higher latencies. They support workloads such as inference, data movement, and replication across regions.

Scale-up networks operate under very different conditions. They connect accelerators within a training cluster and must support latencies on the order of microseconds to a few milliseconds [48]. At this scale, propagation delay

dominates. A 1 ms round-trip budget limits the cluster to a radius of about 50 km.

This makes frontier AI training inherently site-bound. It cannot be distributed across distant data centers, no matter how much bandwidth exists between them. Instead, it requires tightly integrated environments where compute, power, cooling, and ultra-low-latency optical fabrics are co-located [13].

The sovereignty implication is straightforward. A region that cannot build and operate such a scale-up environment cannot host frontier-level training, regardless of its access to hardware or energy. Industry analyses show that without improvements in optical interconnect efficiency, systems remain constrained by short-reach technologies, which in turn drive extreme rack-level power densities and complicate cooling [36, 55]. This creates a concentration effect, where only a few locations can meet all requirements at once.

In practical terms, the physics of latency forces training clusters into tightly bounded geographic areas. If those conditions cannot be met locally, sovereignty over frontier training is out of reach, even if legal and regulatory conditions are satisfied.

4.3 Network layers and their sovereignty roles

Figure 3 shows how different layers of the optical network shape the operational footprint of AI systems.

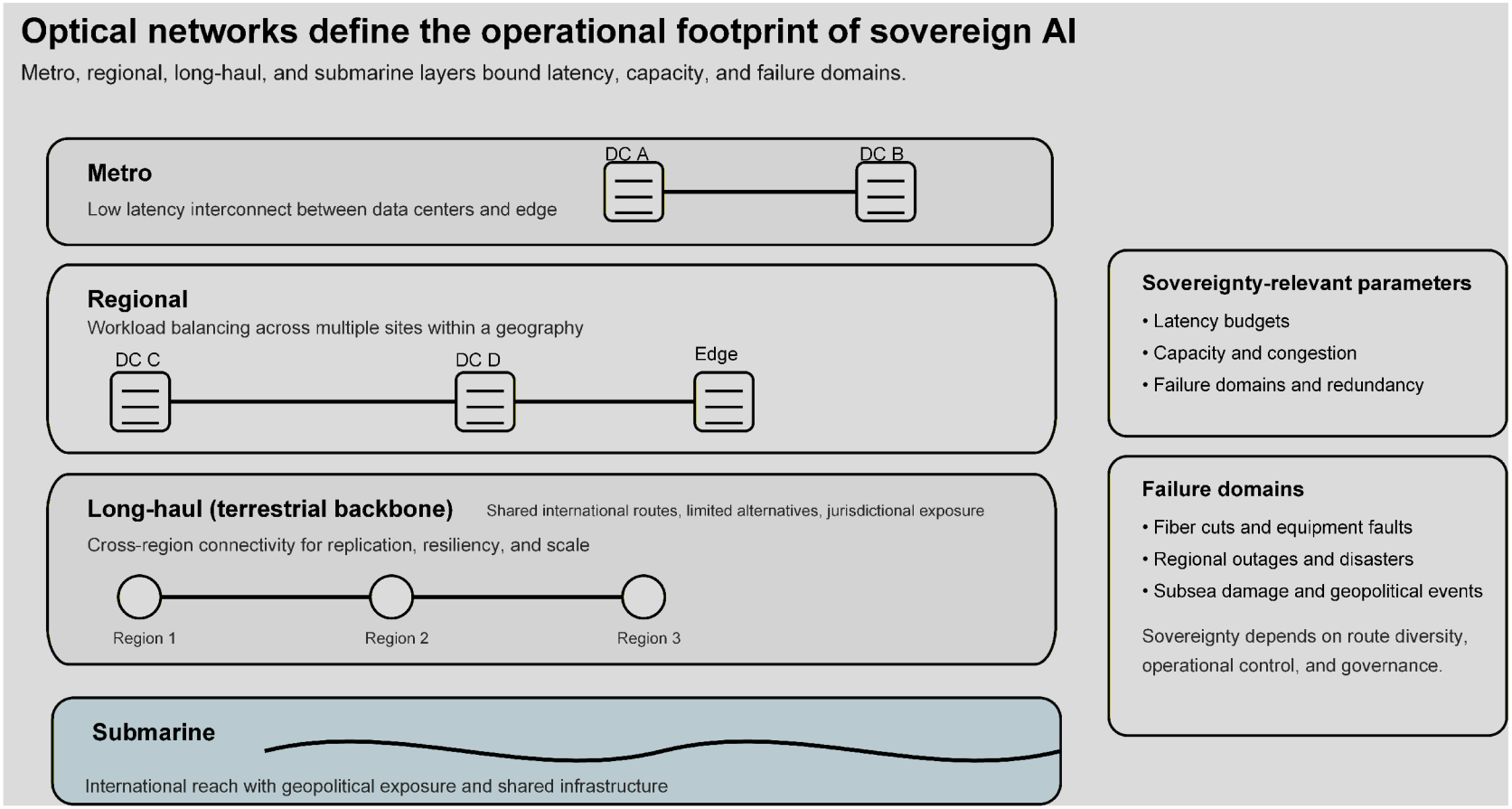


Figure 3: Optical networks as the backbone of AI sovereignty. Metro, regional, long-haul, and submarine layers each impose distinct latency, capacity, and governance constraints that together define the operational footprint of distributed AI infrastructure.

At the **metro level**, networks provide low-latency connectivity within cities, supporting distributed inference and real-time applications. This layer is typically the most controllable from a sovereignty perspective, as it is often owned or regulated domestically.

At the **regional level**, networks connect multiple sites, enabling redundancy and workload shifting. This is where sustainability strategies come into play, allowing workloads to move between locations with different energy or carbon profiles.

At the **long-haul level**, networks connect regions and countries. Latency increases significantly, making these links unsuitable for real-time applications but still useful for bulk data movement and replication. Here, sovereignty concerns shift toward jurisdiction and control, as traffic often crosses multiple legal domains.

Submarine cables represent the most concentrated point of risk. They carry large volumes of traffic across limited physical routes and are difficult to repair. Disruptions can last weeks, and control over landing points can directly affect access. For regions dependent on these links, this creates a structural vulnerability that cannot be mitigated through compute or software alone.

4.4 Jurisdictional control and the role of operators

Optical networks also play a direct role in enforcing data sovereignty. The physical route taken by data determines which legal frameworks apply while it is in transit. Even if compute and storage remain within national borders,

routing traffic through external networks can expose it to foreign regulation or interception [1].

This highlights the difference between legal and operational sovereignty. Legal sovereignty focuses on compliance and data location. Operational sovereignty depends on control over how data actually moves. Optical routing is central to this distinction.

Because of this, routing policies and interconnection decisions are not just technical details. They are sovereignty tools. Regions that control their own infrastructure and routing policies retain more control over data in transit than those that depend on external networks [30].

Telecommunications operators are key in this context. They already manage the infrastructure that determines routing, interconnection, and policy enforcement. This places them in a central role in sovereign AI systems, not just as service providers, but as operators of critical control points [35].

4.5 Optical reach as a sovereignty multiplier

Optical reach directly influences how effectively AI resources can be used. Dense and well-managed networks allow operators to share capacity across sites, adapt to changing conditions, and maintain resilience [21, 18]. Sparse or fragmented networks limit flexibility and increase vulnerability to local failures.

This relationship is not linear. A region with moderate compute but strong connectivity can be more effective than one with more compute but weaker networks, because it can coordinate and adapt its resources more efficiently.

Figure 4 illustrates a key tradeoff between carbon intensity and latency. Cleaner energy is often located far from demand, while latency constraints limit how far certain workloads can be placed. The result is that different workloads have different degrees of flexibility.

Training workloads, which require very low latency, must remain close to compute and cannot take advantage of distant low-carbon sites [13, 69]. Inference workloads have more flexibility but are still constrained by latency. Only batch workloads can be freely moved to optimize for energy or sustainability [33].

This means that sustainability-aware placement is not uniform across workloads. It is limited by physics and workload requirements, and must be treated as a structured constraint rather than a general optimization strategy.

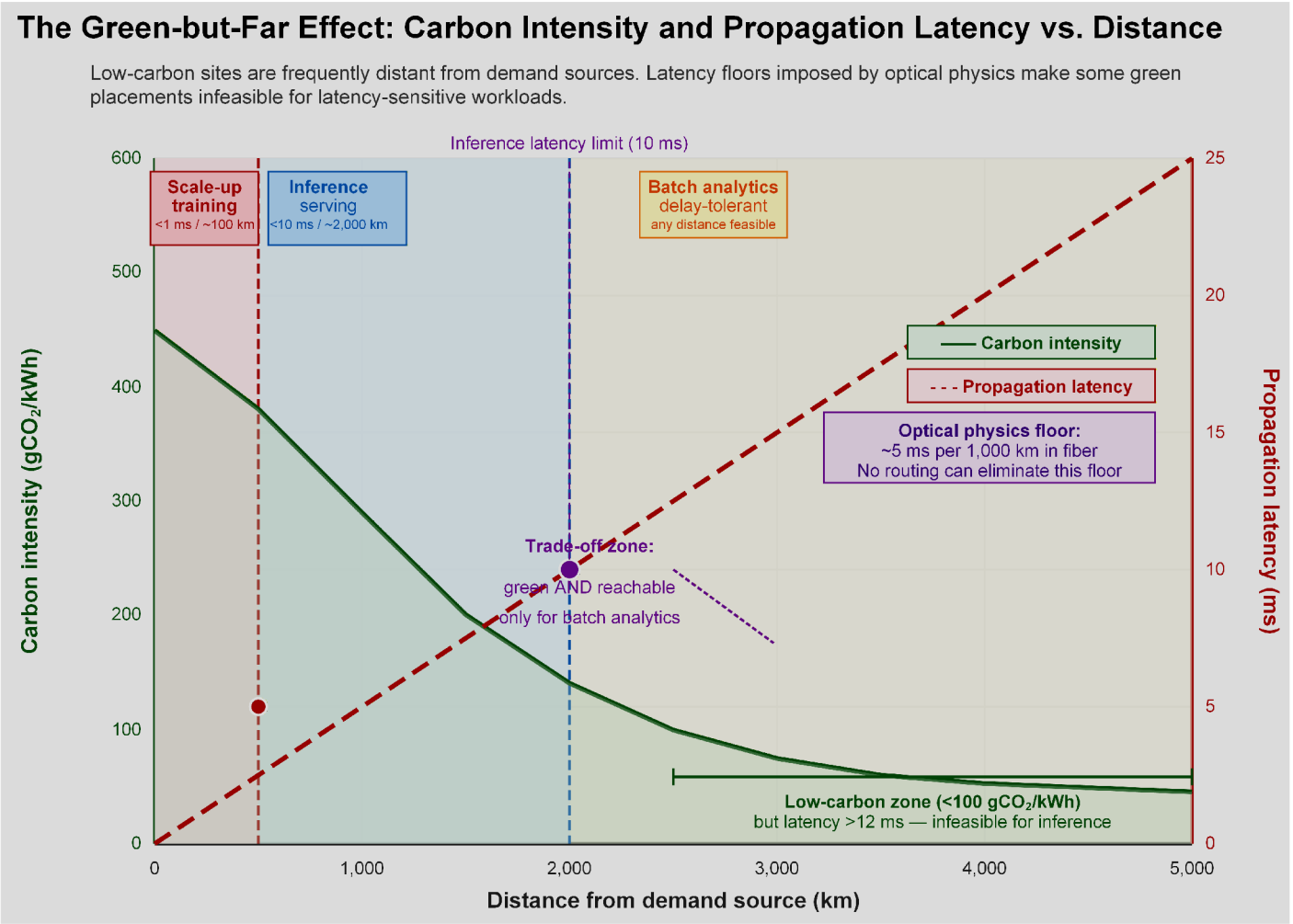


Figure 4: The green-but-far effect: carbon intensity and latency as competing placement constraints across AI workloads.

5 Telemetry as the Foundation of Control and Visibility

As AI infrastructure grows in scale and coupling, effective control increasingly depends on continuous and accurate visibility across compute, network, and energy layers. Telemetry provides this foundation by exposing the real-time state of data centers, optical networks, and supporting energy systems, transforming physical infrastructure from an opaque dependency into an observable and manageable system [20, 7]. In the absence of telemetry, operators are forced to rely on static models, coarse assumptions, or delayed alarms — approaches that are ill-suited to environments characterized by rapid load variation, tight interdependencies, and narrow operating margins. This section argues that telemetry is not merely an operational convenience but a structural prerequisite for sovereign AI control: without the ability to observe infrastructure state in real time, the ability to enforce policies, respect sustainability limits, and retain operational autonomy is nominal rather than effective.

5.1 What telemetry measures and why cross-layer fusion matters

Telemetry in AI infrastructure spans four physically distinct domains, each with its own instrumentation ecosystem, update rate, and schema conventions. In AI-oriented data centers, compute and power telemetry includes rack- and cluster-level power draw, accelerator utilization, thermal gradients across chip and coolant boundaries, and UPS state [62]. Cooling telemetry captures coolant supply and return temperatures, flow rates, make-up water consumption, and ambient wet-bulb temperature, which together determine evaporative cooling efficiency and water permit exposure. In optical networks, telemetry captures signal quality indicators including optical signal-to-noise ratio and pre-FEC bit error rate, spectrum occupancy, per-link latency, and fault and alarm state [20]. Grid sustainability telemetry adds marginal carbon intensity, energy headroom, and on-site generation and storage state, sourced from grid signal providers and building energy management systems.

Collecting these streams in isolation provides limited value. The operational significance of telemetry emerges from cross-layer fusion: combining measurements across domains to reveal causal chains that remain invisible within siloed monitoring systems [32]. Consider three examples. First, a thermal event at a specific rack — detected through coolant return temperature rising above a threshold — may require workload migration to a cooler site, which increases inter-site traffic, which in turn loads optical links that were near congestion. A monitoring system that observes only compute or only network state cannot anticipate this cascade; a cross-layer system can detect the thermal precursor and pre-emptively reserve routing capacity before the migration is triggered. Second, a spike in grid carbon intensity at one site, if detected alongside available power headroom at a low-carbon alternative site and sufficient optical capacity between them, creates an actionable window for sustainability-driven workload shifting that no single-domain monitoring system can identify. Third, a fiber cut that reduces available optical capacity between two sites may force workloads to remain at high-carbon or water-stressed sites that would otherwise have been vacated, because the alternative routing path no longer exists. Cross-layer telemetry makes this constraint visible before a placement decision is made on incorrect assumptions about network availability.

5.2 Streaming architectures and the standardization challenge

Making telemetry operationally useful requires streaming architectures that push measurements continuously into monitoring and control platforms rather than relying on periodic polling or coarse-grained summaries [20, 70]. Polling intervals of minutes or hours are incompatible with AI environments where power draw and thermal conditions change on sub-second timescales during collective communication operations [14]. Streaming telemetry enables near-real-time observability in which changes in load, failures, or environmental conditions are detected within the timescales at which corrective actions remain effective, allowing automation systems and agents to adjust configurations before local degradations escalate into system-wide disruptions.

The standardization challenge underlying this shift is significant and constitutes a practical sovereignty concern in its own right. Each physical domain exposes telemetry through a different protocol stack, schema convention, and transport mechanism, reflecting the fact that these domains evolved independently under separate engineering communities and standards bodies. Integrating them into a unified observability plane therefore requires explicit normalization work that cannot be delegated to any single vendor or platform.

Optical networks are the most mature domain from a telemetry standardization perspective. They increasingly support OpenConfig YANG models — a vendor-neutral, open-source data modeling framework that defines device configuration and state in a structured, machine-readable schema [50] — streamed over gNMI (gRPC Network Management Interface), a high-rate, bidirectional telemetry transport built on gRPC [60]. gRPC itself operates over HTTP/2 and uses Protocol Buffers for compact binary encoding, enabling sub-second streaming of optical KPIs such as signal-to-noise ratio (OSNR), bit error rate (BER), and per-channel power from reconfigurable optical add-drop multiplexers (ROADMs), transponders, and amplifiers directly into control platforms [20]. Despite this progress, vendor implementations vary significantly in schema coverage and update rate fidelity, and many deployed devices still expose only NETCONF [23] or Simple Network Management Protocol (SNMP) [11] interfaces that require polling rather than streaming.

Compute and power domains use a different and less unified set of interfaces. IPMI (Intelligent Platform Management Interface) is a hardware-level specification that allows out-of-band monitoring and management of server hardware — including power draw, thermal state, and hardware status — independently of the operating system [37]. Redfish [22] is a more modern REST-based standard developed by the DMTF (Distributed Management Task Force) that replaces IPMI for server management, providing structured JSON-based access to power consumption, thermal margins, and hardware inventory through a well-defined API. BMC APIs (Baseboard

Management Controller APIs) are vendor-specific interfaces exposed by the embedded management processor present on most server motherboards; they provide low-level hardware telemetry but differ significantly across manufacturers in schema, update rate, and supported metrics, complicating cross-vendor normalization [22]. In practice, Redfish is preferred where available because it provides freshness metadata and structured schemas compatible with automated ingestion; IPMI and proprietary BMC APIs require additional normalization before their outputs can be admitted to a sovereignty-aware control system.

Cooling and building management systems operate under a further distinct set of protocols designed for facilities engineering rather than real-time infrastructure control. BACnet (Building Automation and Control Networks, standardized as ASHRAE 135 and ISO 16484-5) is the dominant protocol for building automation systems, covering HVAC, cooling, lighting, and power distribution; it supports both IP-based (BACnet/IP) and legacy serial (BACnet MS/TP) transport, with read and write operations organized around a structured object model [5]. Modbus is an older serial communication protocol widely used in industrial control and facilities management for reading sensor data from cooling equipment, power distribution units, and environmental monitors; it is simple and robust but lacks structured schemas, requiring manual mapping of register addresses to physical measurements [46]. Neither BACnet nor Modbus was designed for the sub-second update rates or the time-aligned, freshness-certified streams that real-time AI infrastructure control requires, and both need adaptation layers before their outputs can be fused with optical and compute telemetry.

Grid sustainability signals introduce a fourth protocol ecosystem governed by commercial APIs rather than open engineering standards. WattTime [17] is a service that provides real-time and forecast marginal carbon intensity signals — the emissions rate of the next unit of electricity drawn from the grid at a given location and time — sourced from grid operator dispatch data; it is used to determine whether a workload placement decision would

consume low-carbon or high-carbon electricity at execution time. Electricity Maps provides similar signals with broader geographic coverage and both real-time and historical data access. Both services expose REST APIs with their own authentication schemes, rate limits, schema conventions, and update intervals (typically five minutes for real-time signals), and neither was designed with the freshness certification or provenance tracking requirements of a sovereign control system in mind. Grid operator APIs, where available, can provide more authoritative signals but vary significantly by jurisdiction in availability, access terms, and data format [17, 31].

The result is a heterogeneous telemetry landscape in which four physically distinct domains — optical networking, compute and power, cooling and facilities, and grid sustainability — each speaks a different protocol, operates on a different update cadence, and exposes data in a different schema. Integrating these streams into the unified state representation $\theta(t)$ required by the agentic control system is not a commodity integration task but a sovereignty-relevant engineering challenge: an operator who delegates this integration layer to a vendor-supplied management platform or a cloud-based observability service has effectively delegated part of their situational awareness, and therefore part of their operational sovereignty, to an external actor whose behavior and data policies they do not fully govern. The ingestion layer that performs this integration — schema normalization, timestamp alignment, freshness validation, and confidence scoring — is examined in architectural detail in Section 7.

5.3 Data quality, freshness, and AI-specific failure modes

The effectiveness of telemetry depends critically on data quality, freshness, and synchronization, and the failure modes in AI environments are more consequential than in conventional data center operations [20]. Three failure modes deserve explicit treatment.

Stale carbon intensity signals are among the most operationally significant. Grid carbon intensity can change substantially within a single optimization interval — shifting from below to above a regulatory threshold, or vice versa, within minutes during generation dispatch transitions. A placement decision made on a carbon intensity reading that is 15 minutes old may allocate a workload to a site whose grid has since exceeded the operator's carbon policy limit. Because the hard-constraint model eliminates non-compliant configurations entirely rather than penalizing them gradually, this staleness does not produce a slightly suboptimal outcome: it produces a constraint violation that may have regulatory consequences. Freshness thresholds for carbon intensity signals must therefore be set relative to the rate of change of grid dispatch, not relative to the convenience of the data provider's update schedule.

Misaligned timestamps across domains corrupt cross-layer causal inference. Power meters may update every 30 seconds, carbon intensity signals every 5 minutes, water consumption readings every hour, and optical performance monitoring every few seconds. If these streams are fused without explicit timestamp alignment, a power spike that occurred at 14:03 may be correlated with a carbon intensity reading from 13:58 and a water consumption reading from 14:00, producing a misleading picture of the system state at any given moment. In AI training environments where collective communication produces power spikes lasting milliseconds to seconds [14], misalignment of even a few seconds can cause the monitoring system to associate a thermal event with the wrong workload phase, leading to incorrect fault attribution and misdirected corrective action.

Missing or degraded telemetry from critical paths requires a disciplined degradation policy rather than silent substitution with last-known-good values. When alarm state on an optical link is unavailable because a management plane fault has interrupted the telemetry stream, silently assuming the link is healthy can cause the placement optimizer to route traffic over a link that is in active protection switching, producing an actual outage from an

attempted optimization. A principled degradation policy distinguishes between parameters for which conservative substitution is safe — power headroom can be replaced by a lower bound, slowly varying water consumption by the most recent reading with a widened uncertainty interval — and parameters for which no safe substitute exists and optimization should be held until fresh telemetry is restored.

5.4 Telemetry as a sovereignty boundary

As AI infrastructure spans multiple operators, vendors, and jurisdictions, telemetry itself becomes a sovereignty boundary whose governance requires as much attention as the physical infrastructure it monitors. Automation and agentic control across domain boundaries require sharing fine-grained telemetry — power state, network topology, workload placement, carbon intensity — with counterparts in adjacent domains. Unrestricted sharing of this data, however, exposes sensitive operational information: traffic matrices reveal workload patterns and customer behavior; power draw profiles reveal training schedules and model sizes; fault histories reveal infrastructure vulnerabilities. In environments where AI infrastructure is operated by or on behalf of government agencies, critical services, or strategically sensitive industries, this exposure is a direct sovereignty risk.

Recent work on governed telemetry data sharing shows that cross-domain automation and sovereignty can be reconciled through policy-enforced access controls rather than through isolation [45]. By attaching usage policies directly to telemetry streams — specifying which stakeholders can access which measurements, under what time constraints, at what geographic granularity, and for what automated purposes — it becomes possible to support cross-domain control while retaining operational authority and regulatory compliance. A telemetry stream that exposes aggregate carbon intensity at a regional level for sustainability scheduling, while withholding site-level power draw and workload composition from external systems, enables cooperation without disclosure. Enforcing these

policies at the stream level rather than through post-hoc access controls ensures that sovereignty constraints are embedded in the infrastructure workflow rather than dependent on contractual agreements between operators.

This approach has a practical implication for architecture: telemetry governance must be treated as a first-class design concern, not an afterthought applied to an already-designed monitoring system. An organization that designs its telemetry pipeline without sovereignty constraints and then attempts to add them later will find that the pipeline's schema, transport protocols, and aggregation logic have already embedded assumptions about data sharing that are difficult to retrofit. Sovereignty-preserving telemetry requires that access policy, data minimization, and jurisdictional routing be specified at the same time as measurement schema and update rate.

To state this plainly for non-specialist readers: telemetry is not merely technical data about infrastructure operation — it is also information about how and when systems are used, where workloads run, and how resources are consumed. If shared without governance, it reveals sensitive operational details. Treating access policy, data minimization, and jurisdictional routing as first-class design requirements — specified alongside measurement schema and update rate — is therefore a sovereignty obligation, not an afterthought.

5.5 Telemetry as the foundation of operational sovereignty

In the framework developed in this tutorial, telemetry bridges the gap between legal sovereignty — which concerns ownership, jurisdiction, and formal control rights — and operational sovereignty, which requires the ability to observe infrastructure state, make decisions based on local conditions, and execute actions within locally defined policies. Legal sovereignty without telemetry is nominal: an operator may formally control AI infrastructure while

lacking the real-time visibility needed to enforce carbon limits, detect constraint violations, or respond to failures before they propagate. Operational sovereignty requires, at minimum, that the operator can see what the infrastructure is doing at the timescale at which it changes.

Telemetry also strengthens sovereignty in heterogeneous environments where full asset ownership is not achievable. A multi-vendor AI data center that exposes real-time power, cooling, and network telemetry into a unified control plane can enforce local policies regardless of equipment origin. A facility that relies on foreign-manufactured accelerators but monitors their power draw, thermal state, and utilization locally retains operational authority over how those accelerators are used, even when it does not control their design or supply chain. This shifts the locus of sovereignty from asset ownership — an increasingly unrealistic standard given global supply chains — to operational capability: the ability to observe, decide, and act within locally defined constraints [15].

The full realization of this capability requires that telemetry feeds not only human operators but automated reasoning systems that can act on it at the timescales AI infrastructure demands. The agentic AI systems and digital twins that close this loop are examined in the following section.

6 Agentic AI and Digital Twins for Sovereign Operation

As AI infrastructure spans data centers, optical networks, and energy systems, manual operation no longer scales. The number of interacting variables — power draw, thermal state, grid carbon intensity, optical link utilization, workload placement, water permit exposure — combined with tight timing, safety, and sustainability constraints, ex-

ceeds what human operators can manage reliably in real time [7]. Agentic AI addresses this challenge by structuring control logic into autonomous but bounded agents, each responsible for observing system conditions, reasoning about feasible actions, and proposing decisions within explicitly defined limits [20]. An *agent* in this context is a software component that perceives its environment through telemetry, evaluates options against a policy specification, and proposes actions — but does not execute them unilaterally.

Critically, agentic AI does not replace human authority or policy decision making: it structures the *execution* of decisions under predefined policies, safety constraints, and operational boundaries, making automation auditable and its scope of authority explicit. This section argues that agentic AI and digital twins, grounded in the telemetry infrastructure described in Section 5, are the operational mechanisms through which legal sovereignty is converted into genuine operational sovereignty.

6.1 Agent architecture and coordination

Agent-based decision making enables coordination across infrastructure layers without centralizing all control logic in a single controller [20, 70]. Figure 5 illustrates the full closed-loop architecture developed in this section.

To make this flow explicit, the architecture can be read from left to right as a sequence of transformations from observation to validated action. Three distinct telemetry sources form the input layer. **Infrastructure telemetry** captures compute, power, and cooling state, including rack power draw, thermal conditions, and hardware status through interfaces such as Redfish and IPMI [22]. **Network telemetry** captures optical-layer and transport metrics, including latency, faults, and physical-layer indicators such as OSNR, BER, and spectrum utilization via streaming

interfaces such as gNMI and OpenConfig models [60]. **Sustainability telemetry** captures carbon intensity, water utilization, and energy headroom, typically sourced from external grid interfaces and required to meet strict freshness bounds for decision making [17, 31].

These streams are ingested into the **telemetry layer**, which constructs the unified system state $\theta(t)$ through a series of processing steps: streaming and normalization of heterogeneous inputs, timestamp alignment to a common time base, freshness certification to bound decision risk, provenance tracking to preserve source traceability, and cross-domain fusion to produce a consistent multi-layer representation of infrastructure state. These mechanisms are typically implemented using streaming and observability frameworks such as Kafka and OpenTelemetry [58, 51].

The unified state feeds the **agentic AI layer**, organized into two tiers. In the **Tier 1 domain agent layer**, specialized agents perform bounded reasoning within their respective domains: compute placement (workload allocation and migration), power management (grid interaction and headroom control), cooling control (thermal and water system actuation), and optical routing (path selection and capacity management). Each agent proposes candidate actions together with an explicit account of the constraints it satisfies and the trade-offs it introduces.

These proposals are evaluated by the **Tier 2 coordination layer**, which performs joint feasibility checking across domains using policy frameworks such as TM Forum IG1230 and ETSI ZSM [70] and graph-based agent orches-

tration frameworks. Conflicting proposals are not silently resolved: when no feasible joint action exists, the system produces a structured infeasibility report and escalates to the human operator.

The **LLM layer** operates strictly in an advisory and interpretive role, translating human intent into structured objectives and explaining decisions, consistent with prior work on LLM-assisted network automation [19]. All LLM-generated outputs are subject to coordination-layer constraint enforcement and digital twin validation before execution.

Candidate actions are then evaluated by the **digital twin**, which performs pre-execution validation across physical dynamics, model abstraction gaps, and cross-domain interactions, as commonly adopted in digital twin frameworks for infrastructure systems [20]. Only actions that pass this validation stage receive a *validation certificate* and are allowed to reach execution interfaces.

When no valid action can be constructed, control passes to the **human operator**, who defines the sovereignty boundary of the system. The operator receives a structured description of constraint conflicts and retains authority to relax or prioritize constraints, ensuring that policy decisions remain explicitly human-controlled.

Two feedback paths close the loop. **Feedback A** carries measured outcomes back to the telemetry layer to update $\theta(t)$ and confirm action effects. **Feedback B** carries residuals between predicted and observed outcomes exclu-

sively to the digital twin, enabling model recalibration and drift detection. These two feedback paths serve distinct functions and must remain strictly separated.

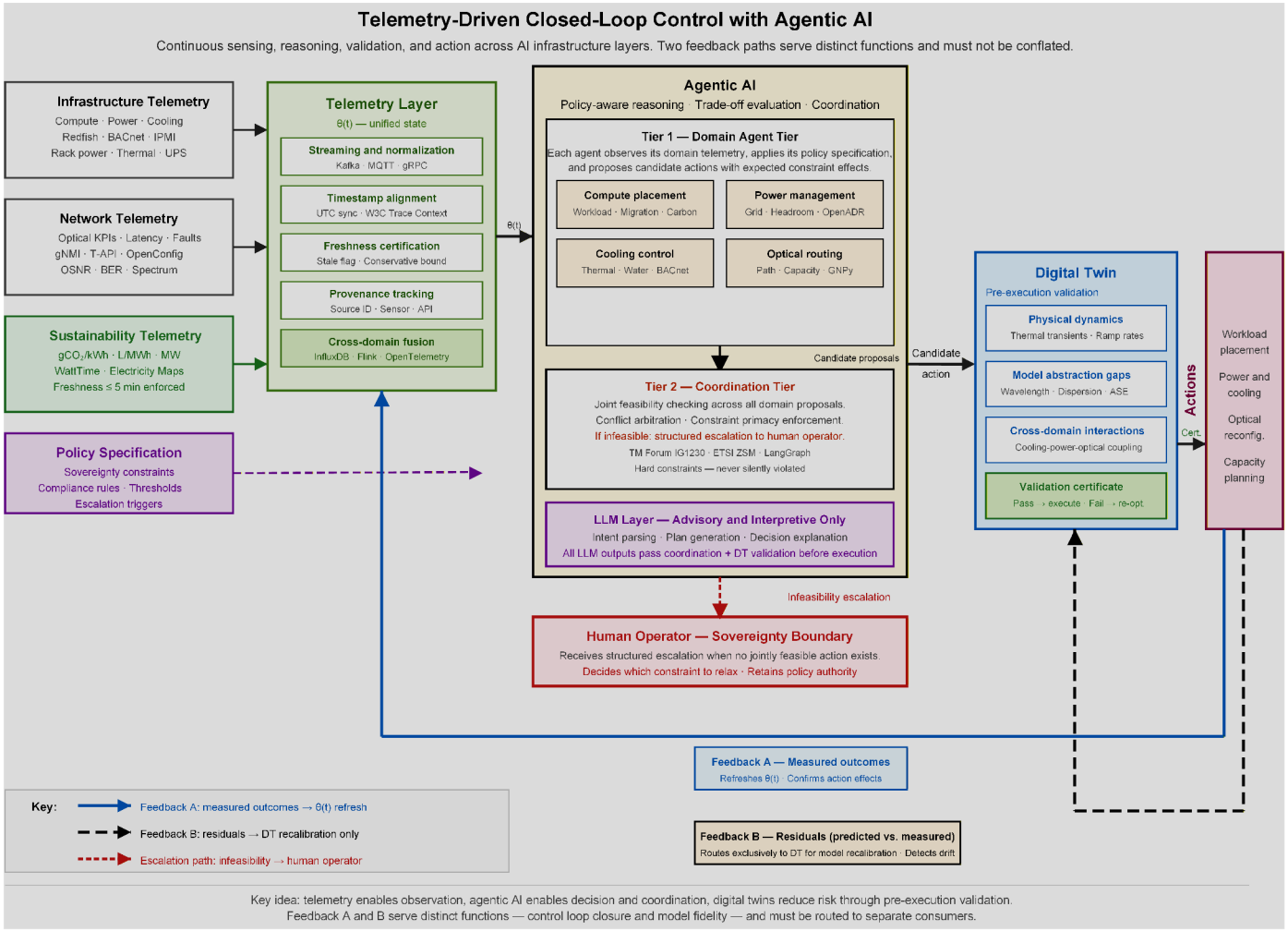


Figure 5: **Telemetry-driven closed-loop control with agentic AI for sovereign infrastructure operation.** Three telemetry streams — infrastructure, network, and sustainability — feed the observability layer, which assembles the unified state representation $\theta(t)$ through streaming normalization, timestamp alignment, freshness certification, and cross-domain fusion. A locally defined policy specification bounds agent authority throughout. The agentic AI block

organizes reasoning into two tiers: a domain agent tier (compute placement, power management, cooling control, optical routing) that proposes candidate actions within bounded scopes, and a coordination tier that evaluates joint feasibility across all proposals. When no jointly feasible action exists, the coordination tier escalates to the human operator (red dashed path) with a structured description of the binding constraint conflict — a sovereignty mechanism that prevents silent policy violation. The LLM layer operates in an advisory and interpretive role only; all outputs are validated before execution. The digital twin validates candidate actions against physical dynamics, model abstraction gaps, and cross-domain interactions before issuing a validation certificate. Only certificate-bearing actions reach southbound interfaces. **Feedback A** (solid blue) returns measured outcomes to the telemetry layer to refresh $\theta(t)$. **Feedback B** (dashed orange) returns prediction residuals exclusively to the digital twin for model recalibration and drift detection. These two paths serve distinct functions — control loop closure and model fidelity — and must not be conflated.

6.2 Digital twins: pre-execution validation and model fidelity

Digital twins play a critical role in validating agent decisions before they affect live infrastructure [20]. A digital twin maintains a synchronized computational model of the physical infrastructure — compute sites, power delivery systems, cooling circuits, and optical network topology — and uses it to simulate the expected consequences of a proposed action before that action is executed. This pre-execution validation is essential in tightly coupled environments where missteps propagate rapidly: a workload migration that overloads a cooling circuit can trigger thermal throttling across an entire rack; a routing change that exceeds optical link capacity can degrade signal quality below the FEC threshold and cause a link failure; a power allocation change that violates a demand response contract can impose financial penalties and trigger grid-side protection actions.

The digital twin operates on three classes of constraints that the optimization model alone cannot fully enforce. The optimization layer is formulated as a mixed-integer linear program (MILP), which represents the system using linear constraints and discrete decision variables to enable tractable, global optimization [8]. While this formulation is effective for capturing steady-state feasibility and coordinating decisions across domains, it necessarily abstracts away non-linear dynamics, transient effects, and higher-order physical interactions.

First, **physical dynamics** that are abstracted away in the MILP formulation: thermal transients, power ramp rates, coolant flow stabilization times, and optical amplifier gain settling times all evolve on timescales shorter than the optimization cycle and can cause constraint violations during the transition to a new configuration even when the steady-state target is compliant. Second, **model abstraction gaps**: the optimization graph represents the optical network as a set of logical edges with capacity and delay parameters, but the physical layer has wavelength assignments, amplifier cascades, and chromatic dispersion margins that determine whether a proposed routing is physically realizable at the optical layer. Third, **cross-domain interactions** that are too complex to enumerate as MILP constraints: the interaction between a specific cooling configuration and the ambient temperature forecast at a given site, for example, may determine whether a proposed power increase is thermally safe, but encoding this interaction as a linear constraint requires approximations that the digital twin can evaluate more faithfully using a higher-fidelity thermal model.

Maintaining the fidelity of the digital twin over time is as important as its initial accuracy. Physical infrastructure changes — equipment replacements, cooling system upgrades, fiber route additions, grid connection modifications — must be reflected in the twin promptly, or the twin will validate actions against a model of infrastructure that no longer exists. Model drift, in which the twin's representation progressively diverges from physical reality, is a failure mode with direct sovereignty consequences: an operator who believes their digital twin provides accurate pre-exe-

cution validation but whose twin has drifted is making decisions on the basis of an inaccurate model without knowing it. Continuous reconciliation between twin predictions and measured telemetry outcomes, flagging cases where the twin predicted one outcome and the infrastructure produced another, is the primary mechanism for detecting and correcting model drift [20, 40].

6.3 Large language models at the human-automation interface

Large language models contribute at the interface between human intent and automated control, occupying a specific and bounded role that is distinct from both policy specification and action execution [19]. Operators typically express operational goals in abstract terms: reduce the carbon footprint of the training cluster, maintain inference latency below 50 ms during the maintenance window, or prepare the facility for a predicted grid stress event at 18:00. These expressions are underspecified from the perspective of the agentic control system: they do not identify which workloads to migrate, which sites are eligible, which constraints take precedence when trade-offs arise, or what the acceptable degradation in secondary objectives is.

Large language models (LLMs) bridge this gap by translating natural language intent into structured objective specifications and constraint parameterizations that the agentic system can evaluate against real-time telemetry and digital twin models [20]. They also enhance operational transparency by generating human-readable explanations of decisions: why a specific workload was placed at a specific site, which constraints were binding, what trade-offs were made between carbon and latency, and what alternative actions were considered and rejected. This explanatory capability is particularly important for sovereignty, because it makes automated decisions au-

ditable: an operator can reconstruct the reasoning behind a configuration change and verify that it was consistent with declared policy.

The boundary around LLM involvement must be drawn carefully. LLMs are probabilistic systems whose outputs are not guaranteed to be correct, consistent, or policy-compliant. Allowing LLM outputs to drive infrastructure actions directly — without validation through the deterministic constraint-checking of the agentic system and the pre-execution simulation of the digital twin — introduces a failure mode in which plausible-sounding but incorrect instructions are executed on live infrastructure [19]. The appropriate role is therefore advisory and interpretive: LLMs propose structured objectives and explain decisions, but every proposed action passes through the agent coordination layer and digital twin validation before execution. This architecture preserves the intelligibility benefits of natural language interaction while containing the risk of LLM error within the safety envelope provided by deterministic validation.

6.4 From manual operation to sovereign autonomous control

The transition from manual operation to agentic control represents a qualitative change in the nature of operational sovereignty, not merely an increase in automation level. Table 3 characterizes this transition across five dimensions: observability, pre-execution validation, decision autonomy, human oversight requirement, and primary failure mode. The comparison shows that higher autonomy is not achieved by reducing human oversight but by grounding decisions in continuous telemetry, validating them against physical models, and constraining their scope within explicitly specified policies. The primary failure mode shifts from human error and reaction-time limits under manual operation, to rule brittleness and edge-case blindness under rule-based automation, to model drift and coordination

failure under agentic control — a progression that reflects increasing capability alongside shifting risk profiles that require different governance responses.

Table 3: Comparison of control approaches for AI infrastructure sovereignty. Higher autonomy shifts the primary failure mode from human reaction limits to model fidelity and coordination, requiring governance responses focused on twin reconciliation and escalation path design rather than operator training.

Approach	Observability	Validation	Autonomy	Human oversight	Primary failure mode
Manual operations	Low, periodic	None	Low	Continuous	Reaction time; fatigue
Rule-based automation	Medium, polled	Rule-check only	Medium	Exception-driven	Rule brittleness; edge cases
Agentic with digital twins	High, streaming	Pre-execution simulation	High	Escalation-driven	Model drift; coordination failure

Entries are qualitative characterizations of typical deployments within each approach; actual observability, validation depth, and oversight requirements vary with implementation scale and policy specification.

6.5 Agentic control as operational sovereignty

As defined in Section 1, this work distinguishes between *legal sovereignty* and *operational sovereignty*. Legal sovereignty refers to formal control over ownership, jurisdiction, and access rights, while operational sovereignty denotes the ability to observe, decide, and act on infrastructure state in practice. In the framework developed in this tutorial, agentic AI and digital twins are the mechanisms that transform legal sovereignty into operational sovereignty.

Legal sovereignty alone does not guarantee the ability to observe infrastructure state, enforce sustainability constraints, or respond to failures in real time. An operator may formally own AI infrastructure and therefore satisfy legal sovereignty, yet lack operational sovereignty if they depend on external systems or providers to monitor, validate, or execute control actions. In such cases, control exists at the level of policy but not at the level of execution.

Agentic control closes this gap by establishing operational sovereignty through locally grounded control loops. Decisions are derived from locally collected telemetry rather than vendor-supplied state estimates. Candidate actions are validated against locally maintained digital twin models rather than external simulation services. Executed actions are constrained by locally defined policies rather than platform defaults. When automated competence reaches its limit, the system escalates to local human authority with structured explanations rather than relying on external support [49]. This combination — local observation, local validation, local policy enforcement, and structured escalation — constitutes operational sovereignty in practice.

This interpretation of operational sovereignty aligns with policy-aware agentic control frameworks developed in adjacent domains, including cloud data pipeline governance, intelligent infrastructure monitoring, and smart grid control [41, 40]. The following section integrates the physical infrastructure examined in Sections 3 and 4, the telemetry foundation of Section 5, and the agentic control architecture developed here into a unified reference architecture for sovereign AI infrastructure.

7 Reference Architecture for AI Infrastructure Sovereignty

The preceding sections have examined the physical, operational, and control dimensions of AI infrastructure sovereignty in sequence: data center sustainability limits, optical network boundaries, telemetry as the observability foundation, and agentic AI with digital twins as the control mechanism. This section integrates these elements into a unified reference architecture that specifies not only what components are required but how they connect, what flows across their interfaces, how the system degrades gracefully under partial failure, and how sovereignty level can be assessed against a structured framework. The architecture is not prescriptive of specific vendors or implementations; it is prescriptive of functional responsibilities, interface contracts, and sovereignty properties that any compliant implementation must satisfy.

7.1 Architectural overview and design principles

Figure 6 presents the reference architecture as a closed-loop control system organized across four functional layers. The layers are vertically integrated through defined interfaces and horizontally coupled through the shared telemetry plane that makes the state of each layer visible to all others. A critical architectural distinction governs the feedback paths that close the loop from Layer 4 back into the system. Two physically separate feedback streams

serve fundamentally different functions and must not be conflated. **Feedback A** (solid blue) carries measured outcomes — actual power draw, carbon consumption, water usage, link utilization, and workload placement observed after execution — back to the observability layer, where they refresh the unified state representation $\theta(t)$ for the next control cycle. This is the primary closed-loop signal confirming whether executed actions produced their intended effects. **Feedback B** (dashed orange) carries residuals — the difference between digital twin predictions and measured outcomes — exclusively to the digital twin for model recalibration. This stream provides the ground-truth signal needed to detect and correct model drift before it compromises validation reliability. Routing Feedback B into the observability layer rather than the digital twin, or conflating it with Feedback A into a single monitoring stream, destroys the model drift detection capability that is essential for maintaining twin fidelity over time. These two streams must be maintained separately and routed to their respective consumers at all times, including during partial failures of the telemetry or execution infrastructure.

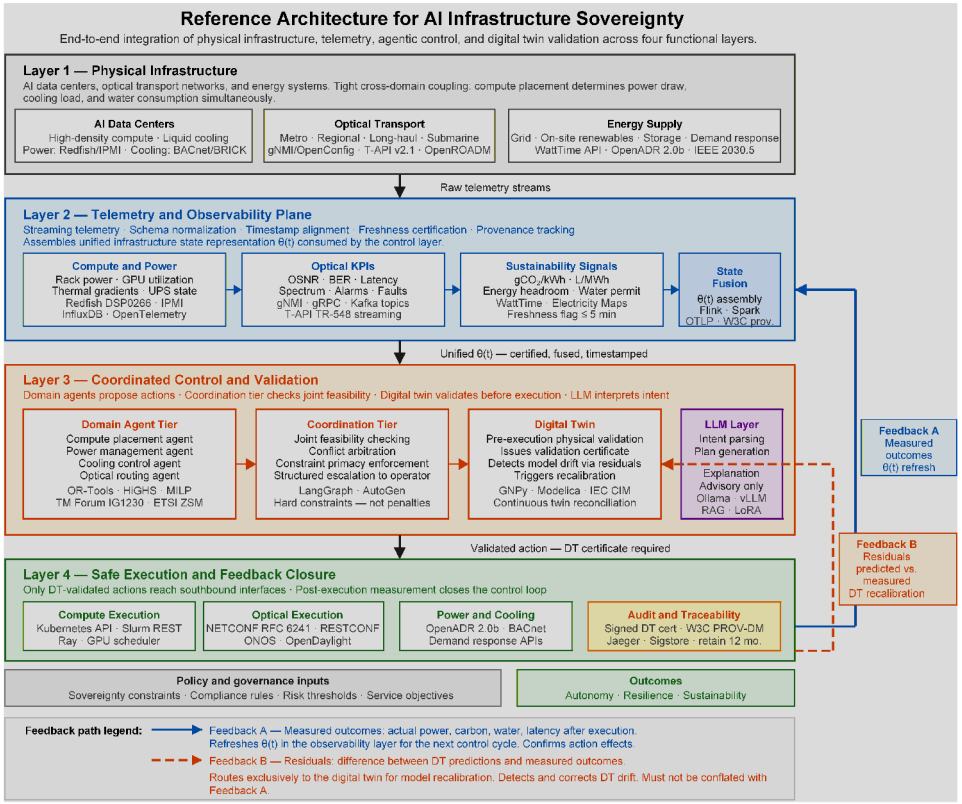


Figure 6: **Reference architecture for AI infrastructure sovereignty, organized across four functional layers.** The physical layer (Layer 1) provides AI data centers, optical transport, and energy systems with tight cross-domain coupling. The observability layer (Layer 2) streams, normalizes, and time-aligns cross-domain telemetry into the unified infrastructure state representation $\theta(t)$. The control layer (Layer 3) applies domain agent reasoning, coordination-tier joint feasibility checking, digital twin pre-execution validation, and LLM-assisted intent parsing to produce safe, policy-compliant candidate actions. The execution layer (Layer 4) applies those actions through domain-specific southbound interfaces and enforces the invariant that only digitally twin-validated actions reach live infrastructure. Two distinct feedback paths close the loop: **Feedback A** (solid blue) returns measured outcomes to the observability layer to refresh $\theta(t)$; **Feedback B** (dashed orange) returns prediction residuals exclusively to the digital

twin for model recalibration and drift detection. These two streams serve different functions — control versus model fidelity — and must not be conflated. Sovereignty properties are realized at each layer and depend on the integrity of the interfaces between them.

Three design principles govern the architecture. The first is *local grounding*: every decision made by the control layer must be based on telemetry collected locally, validated against models maintained locally, and executed within policies defined locally. Dependence on external state estimates, remote simulation services, or platform-provider control logic at any of these steps reduces operational sovereignty proportionally. The second is *constraint primacy*: sustainability and safety constraints are encoded as hard boundaries that eliminate non-compliant actions from the feasible set rather than as penalties that can be traded against performance objectives. This encoding reflects the legal and physical reality that regulatory maxima are not preferences but absolute limits whose violation carries consequences that cannot be offset by gains elsewhere. The third is *graceful degradation*: the architecture must specify safe behaviors for every foreseeable failure mode across its components, ensuring that partial outages produce conservative, bounded responses rather than unsafe or unconstrained ones. An architecture that requires all components to be fully operational in order to behave safely is not suitable for sovereign infrastructure, where resilience under adversarial or degraded conditions is a first-order requirement.

7.2 Layer 1: Physical infrastructure

At the foundation of the architecture lies the physical infrastructure examined in Sections 3 and 4. AI-oriented data centers host high-density compute platforms and advanced cooling systems whose design is governed by the power, thermal, and water constraints analyzed in Section 3. These facilities are interconnected through metro, re-

gional, and long-haul optical networks whose latency floors, capacity ceilings, and failure domains define the feasible geographic scope of sovereign AI operation, as analyzed in Section 4. Energy systems supply power from grid connections, on-site renewable generation, and storage assets, often under tight capacity, carbon intensity, and demand response obligations [38, 42].

The defining characteristic of the physical layer, from an architectural perspective, is tight cross-domain coupling. A decision about compute placement determines power draw at the destination site, which determines cooling load and water consumption, which may activate a water permit constraint, which removes that site from the feasible set for future placements. A change in grid carbon intensity at one site changes the relative sustainability ranking of available placements, which changes traffic flows through the optical network, which changes link utilization and may activate a capacity constraint. These causal chains propagate across domain boundaries on timescales ranging from milliseconds for power dynamics to hours for carbon intensity shifts to seasons for water permit limits. The observability layer must capture all of them; the control layer must reason across all of them simultaneously.

A sovereignty-relevant dependency at the physical layer that deserves explicit architectural attention is the supply chain for next-generation optical components. Sustaining AI scale requires continued advances in photonic integration, transceiver density, and coherent DSP efficiency to improve bandwidth per watt and per unit of rack space [54, 63]. Regions that lack domestic capacity in photonic component design or manufacturing face a structural dependency at the physical layer that propagates upward: even a fully sovereign control architecture cannot compensate for physical layer constraints imposed by supply chain disruption or export restriction on critical optical components. This dependency is analogous to the semiconductor dependency that has already shaped AI sovereignty policy in multiple jurisdictions and warrants similar strategic attention.

7.3 Layer 2: Unified observability

The observability layer is the architectural mechanism through which the physical infrastructure becomes a controllable system. As analyzed in Section 5, it ingests streaming telemetry from four physically distinct domains — compute and power, cooling and water, optical network, and grid sustainability — each with heterogeneous protocols, schemas, and update rates. These domains correspond directly to the telemetry structure shown in Figure 6: compute and power state, optical KPIs, sustainability signals, and the resulting state fusion process that assembles these inputs into the unified representation $\theta(t)$. The layer's functional responsibilities are schema normalization, timestamp alignment, freshness validation, confidence scoring, provenance tracking, and cross-domain fusion, producing a unified infrastructure state representation $\theta(t)$ that the control layer can consume without domain-specific knowledge of the underlying instrumentation.

These functions operate as a processing pipeline rather than independent steps. Streaming ingestion and schema normalization align heterogeneous inputs; timestamp alignment enforces a common temporal reference; freshness certification bounds decision risk; provenance tracking preserves source traceability; and cross-domain fusion produces a consistent multi-layer state representation. The output of this pipeline is a certified, timestamped, and fused state vector that defines the observable boundary of the system.

The interface contract between the observability layer and the control layer specifies three properties that are sovereignty-relevant. First, *completeness*: the state representation must include all parameters that appear as constraint inputs in the control layer's optimization, with no silent gaps that cause the optimizer to evaluate constraints against unobserved or assumed values. Second, *freshness certification*: each parameter in the state representation must carry a timestamp and a freshness flag indicating whether it was measured within its domain-specific maximum age, estimated from a forecast within a specified uncertainty bound, or substituted from a conservative

bound because no fresh measurement was available. The control layer must consume these flags and adjust its behavior accordingly, holding optimization when critical parameters are stale rather than proceeding on potentially invalid state. Third, *provenance*: the origin of each measurement — the specific sensor, API endpoint, or estimation model that produced it — must be traceable through the state representation to support audit and fault attribution.

Vendor neutrality at the observability layer is realized through adherence to open telemetry standards rather than through vendor-specific management APIs. Optical network telemetry should be sourced through OpenConfig YANG models and gNMI streaming over gRPC [60]. Compute and power telemetry should be sourced through Redfish [22] or IPMI interfaces. Grid sustainability signals should be sourced through open or documented APIs with explicit freshness and methodology disclosure [17]. Where vendor implementations deviate from open standards, the observability layer should mediate through normalization adaptors rather than propagating vendor-specific representations into the control layer. This architectural boundary — open standards at the observability interface, proprietary implementations below it — is the mechanism through which the control layer achieves independence from specific equipment vendors without requiring that all physical layer components be replaced with standards-compliant alternatives simultaneously.

7.4 Layer 3: Coordinated control

The control layer realizes the agentic architecture described in Section 6. It consumes the unified state representation from the observability layer, applies domain-specific and coordination-level reasoning to produce a candidate joint action, and submits that action to the digital twin for pre-execution validation before releasing it to the execution layer. As shown in Figure 6, the layer is structured into three primary components — the domain agent tier, the

coordination tier, and the digital twin — complemented by an LLM interface that supports intent interpretation and explanation without participating in execution.

The *domain agent tier* comprises specialized agents for compute placement, power management, cooling control, and optical routing. Each agent consumes the subset of the state representation relevant to its domain, applies its policy specification to generate candidate actions, and exposes those candidates to the coordination layer through a structured proposal interface that specifies the action, its expected effect on relevant state parameters, the constraints it satisfies, and the constraints it may stress in adjacent domains. This proposal interface is the mechanism through which the coordination layer can evaluate joint feasibility without requiring each agent to model the full cross-domain state.

The *coordination tier* receives proposals from all domain agents and applies joint feasibility checking to determine whether a combination of proposals simultaneously satisfies all active constraints across all domains [70]. This function is aligned with cross-domain automation frameworks such as TM Forum IG1230 and ETSI ZSM, in which policy constraints are enforced as hard limits rather than optimization penalties. When a jointly feasible combination exists, it is assembled into a candidate joint action and forwarded to the digital twin. When no jointly feasible combination exists, the coordination tier applies a structured escalation protocol: it identifies the binding constraint or set of constraints responsible for infeasibility, generates a human-readable explanation of the conflict, and holds the system in its current configuration pending operator instruction. This escalation path is a sovereignty mechanism: it ensures that constraint violations are never resolved silently by automated systems making implicit policy decisions, but are always surfaced to the operator with the information needed to make an explicit one.

The *digital twin* validates the candidate joint action against a higher-fidelity model of physical infrastructure than the optimization model alone can provide [20]. As analyzed in Section 6, this includes physical dynamics, model abstraction gaps, and cross-domain interactions that cannot be faithfully encoded as linear constraints in the optimization formulation. The twin returns either a validation certificate — an explicit authorization token required for execution — or a rejection specifying which predicted constraint would be violated, which triggers re-optimization with a tightened feasible set that excludes the rejected action. The validation certificate functions as a control invariant: no action may reach a southbound interface without it. The twin also continuously reconciles its predictions against measured telemetry outcomes, flagging model drift when the physical system's response deviates systematically from prediction and triggering model recalibration before drift accumulates to the point where validation becomes unreliable.

A complementary component of the control layer is the *LLM interface layer*, shown explicitly in Figure 6. This layer operates strictly in an advisory and interpretive role: it translates human intent into structured objectives, supports plan generation, and provides explanations for agent decisions, but it does not participate in constraint enforcement or action execution. All LLM-derived outputs must pass through coordination-layer feasibility checking and digital twin validation before they can influence system behavior.

7.5 Layer 4: Safe execution and feedback closure

The execution layer translates validated actions into configuration changes applied to physical infrastructure through domain-specific southbound interfaces. As shown in Figure 6, these interfaces span four domains: compute execution (e.g., workload scheduler APIs such as Kubernetes or Slurm), optical execution (e.g., NETCONF/RESTCONF interfaces to network controllers), power and cooling control (e.g., demand response APIs

and building management systems), and audit and traceability systems that record all executed actions. The layer enforces a strict invariant: only actions that carry a digital twin validation certificate are submitted to southbound interfaces. Actions that have not passed pre-execution validation, or whose validation certificate has expired because infrastructure state changed between validation and execution, are withheld and returned to the control layer for re-evaluation against the current state.

All executed actions generate signed records that bind the action, the state under which it was validated, and the identity of the validation process. These records form the audit and traceability interface shown in Figure 6, enabling post-hoc verification, compliance auditing, and attribution of responsibility [73]. This mechanism ensures that execution remains accountable and that validated control decisions can be reconstructed and inspected over time.

In sovereign infrastructure deployments, the integrity of the audit trail itself becomes a sovereignty concern. Signed execution records stored in a conventional centralized log are vulnerable to post-hoc modification, selective deletion, or suppression under regulatory or legal pressure from external actors — precisely the scenarios that operational sovereignty is designed to resist. For deployments where audit integrity under adversarial conditions is a requirement, execution records can be anchored to a tamper-resistant trust substrate, such as a permissioned distributed ledger, to provide non-repudiation, provenance integrity, and verifiable lifecycle traceability across infrastructure actions, control decisions, and AI-assisted operations [66, 27].

The architectural role of this mechanism is additive rather than substitutive. A distributed ledger does not replace the local control loop, the digital twin validation process, or the signed execution records described above. It ex-

tends them by providing an independently verifiable commitment to the sequence and content of sovereignty-relevant events: that a specific action was proposed at a specific time, validated against a specific twin state, executed through a specific southbound interface, and recorded without alteration. This commitment is particularly relevant in multi-operator or multi-jurisdictional deployments, where audit records must be accepted as authoritative by parties who did not participate in their creation and who may have adversarial interests in contesting their contents [27].

Permissioned ledger architectures, such as Hyperledger Fabric [2], are better suited to this role than public blockchain designs in most sovereign infrastructure contexts, because they allow membership, write authority, and read access to be governed by the participating operators and regulators rather than by open consensus. The ledger does not need to store full execution payloads — storing cryptographic hashes of signed records together with sufficient metadata to identify the action, the validation certificate reference, and the infrastructure state snapshot is sufficient for non-repudiation while keeping on-chain data volume tractable [66]. This design also decouples audit integrity from the availability of the primary telemetry and control systems: even if an operator's internal logging infrastructure is compromised or unavailable, the on-chain commitment provides an independent reference point for compliance auditing and incident reconstruction.

This mechanism is optional in the reference architecture: deployments that operate within a single administrative domain with strong internal audit controls and no multi-party accountability requirement can satisfy the audit and traceability function entirely through the signed execution records and W3C PROV-DM [47] provenance model described above. The distributed ledger extension is recommended for deployments that cross operator boundaries,

involve regulated workloads subject to external audit, or operate in environments where the integrity of the audit trail may itself be contested.

The feedback path from execution to observability carries two classes of information that serve distinct purposes, corresponding to the two feedback loops in Figure 6. *Feedback A* (*measured outcomes*) — the actual power draw, carbon consumption, water usage, link utilization, and workload placement observed after execution — updates the state representation $\theta(t)$ for the next control cycle. This is the primary closed-loop signal that confirms whether executed actions produced their intended effects or whether corrective action is required.

Feedback B (*residuals*) — the difference between digital twin predictions and measured outcomes — is routed exclusively to the digital twin for model recalibration. This signal enables the detection of systematic deviations between predicted and observed system behavior, triggering model updates before drift compromises validation reliability. These two feedback streams must be maintained strictly separate and routed to their respective consumers; conflating them into a single monitoring stream removes the model drift detection capability that is essential for maintaining twin fidelity over time.

7.6 Graceful degradation under partial failure

A sovereign architecture must specify safe system behavior for every foreseeable component failure, not only for nominal operation. These failure-handling mechanisms preserve operational sovereignty under degraded condi-

tions by ensuring that control decisions remain bounded, auditable, and locally governed. Four failure modes warrant explicit treatment.

When **telemetry is degraded or unavailable** for one or more parameters, the observability layer applies the three-tier freshness policy described in Section 5: forecast substitution with widened uncertainty for slowly varying parameters, conservative bounding for safety-critical parameters, and optimization hold for parameters with no safe substitute. The state representation $\theta(t)$ must reflect these substitutions explicitly through freshness flags, and the control layer must restrict its action space accordingly, avoiding decisions that depend on unverified or stale state.

When the **digital twin is unavailable**, the control layer falls back to a conservative operating mode in which only actions within a pre-specified safe envelope — defined as actions that do not increase power draw, do not migrate workloads, and do not modify routing — are permitted without validation. This envelope is defined at deployment time and reviewed periodically; it represents the operator's judgment about which actions are sufficiently low-risk to execute without pre-execution simulation. Actions outside the safe envelope are held until the twin is restored. The invariant that only validated actions may be executed is relaxed in a controlled and explicitly bounded way, preserving safety while maintaining limited operational continuity.

When the **optimization is infeasible** — when no placement and routing assignment exists that satisfies all active sustainability, latency, and capacity constraints simultaneously — the system interprets infeasibility as a sovereignty signal rather than a solver failure, as analyzed in Section 4. Delay-tolerant workloads are deferred; portable workloads are migrated to the least constrained available site with an explicit constraint relaxation logged

for operator review; non-portable, latency-sensitive workloads are maintained at their current placement with a constraint violation alert escalated to the operator. The system never silently violates a hard constraint to avoid infeasibility, ensuring that policy decisions remain explicit and operator-controlled.

When **agent coordination fails** due to communication loss between domain agents and the coordination tier, each domain agent falls back to its most recent valid local policy, freezing its proposed actions at the last jointly feasible configuration. This freeze-on-failure behavior prevents domain agents from taking unilateral actions that may be locally valid but jointly infeasible, at the cost of reduced responsiveness to infrastructure changes during the coordination outage. This mechanism preserves coordinated control semantics by ensuring that no domain operates outside the last validated global state.

7.7 Sovereignty assessment framework

The reference architecture enables a structured assessment of the sovereignty level achieved by a given deployment, organized across the four layers and the three design principles. As defined in Section 1, this assessment must distinguish between legal sovereignty — formal control over ownership, jurisdiction, and access — and operational sovereignty, which depends on the ability to observe, decide, validate, and execute actions across the infrastructure stack. Table 4 provides a qualitative framework mapping each layer to the sovereignty properties it contributes, the indicators that signal high versus compromised sovereignty at that layer, and the primary risk if those properties are not realized. This framework is intended as a diagnostic instrument for operators and policymakers evaluating existing deployments or specifying requirements for new ones.

Table 4: Sovereignty assessment framework for AI infrastructure deployments. Each architectural layer contributes distinct sovereignty properties; compromised sovereignty at any layer propagates upward, limiting the effectiveness of sovereign control at higher layers regardless of their individual integrity.

Layer	**Sovereignty property**	**High sovereignty indicator**	**Primary risk if compromised**
Physical infrastructure	Local energy, cooling, and optical control	Domestic grid access; on-site generation; owned fiber routes	Environmental limits block deployment; foreign regulatory reach over data in transit
Unified observability	Local, vendor-neutral, governed telemetry	Open standards; freshness-certified state; stream-level access policy	Decisions on stale or externally filtered state; telemetry exposure risk
Coordinated control	Local reasoning, validation, and escalation	Locally maintained twin; structured escalation; no external control dependency	Silent constraint violations; dependence on vendor control plane for policy enforcement
Safe execution	Validated, auditable, reversible actions	Validation certificates enforced; residuals tracked; southbound interface independence	Unvalidated actions on live infrastructure; loss of audit trail

A deployment achieves high sovereignty when each layer satisfies its sovereignty properties and the interfaces between layers enforce the three design principles — local grounding, constraint primacy, and graceful degradation — through explicit control, validation, and data exchange contracts. Partial sovereignty, which is the realistic outcome for most deployments given global supply chain dependencies, is assessed by identifying which layers and which interfaces fall short of full sovereignty and how these deficiencies propagate upward to constrain control at higher layers. This layered assessment enables targeted investment by linking specific infrastructure gaps to their operational consequences: strengthening sovereignty at the physical layer by securing domestic energy and fiber assets, at the observability layer by adopting open telemetry standards and governed data sharing, at the control layer by deploying locally maintained digital twins and structured escalation paths, and at the execution layer by enforcing validation certificates and maintaining southbound interface independence from vendor management platforms.

Table 5 maps each architectural layer to representative open standards and reference tools that enable practical implementation. The table abstracts from specific versions and focuses on the dominant interfaces and frameworks used in current deployments, including OpenConfig/gNMI for telemetry [60], T-API for topology abstraction [56], Redfish for hardware management [22], and OpenTelemetry for observability [51]. Streaming platforms such as Kafka [58] and MQTT [59] and carbon-aware interfaces [17, 31] complete the operational stack. Proprietary substitutions reduce vendor neutrality and must be evaluated for their impact across layers.

Table 5: Implementation reference for sovereign AI infrastructure. Each layer is mapped to key standards and tools. Proprietary substitutions may impact vendor neutrality and sovereignty. Tools listed are representative examples of available frameworks; listings do not constitute endorsement and should be evaluated against current release status before deployment.

Layer	Standards	Tools	Sovereignty note
L1 — Physical			
Optical transport	OpenConfig; gNMI; T-API	GNPy; Telegraf; OpenROADM	Prefer domestic fiber; avoid SNMP
Compute	Redfish; IPMI	OpenBMC; Redfish Python	Prefer Redfish; IPMI lacks freshness
Cooling	BACnet; BRICK	Niagara; BRICK toolkit	Normalize before $\theta(t)$
Energy	WattTime; Elec. Maps; OpenADR	Carbon SDK; PyPSA	Freshness $\leq$ 5 min
L2 — Observability			
Modeling	YANG; OpenConfig; OTLP	Pyang; OTEL Collector	Canonical YANG paths required
Transport	Kafka; MQTT	Kafka; Mosquitto	Replication $\geq$ 3; TLS
Storage	InfluxDB; OpenMetrics; W3C Trace	InfluxDB; Flink	Tag source, time, freshness
L3 — Control			
Optimization	MILP; ACTN; PCE	OR-Tools; HiGHS; GNPy	Hard constraints (carbon, water)

Coordination	TMF AN; ETSI ZSM	Graph-based orchestration frameworks	Deterministic escalation
Digital twin	CIM; OpenDSS; GNPy	GNPy; OpenDSS; Modelica	Continuous reconciliation
LLM	OpenAI API; HF	Ollama; vLLM; LlamaIndex	Advisory role only
L4 — Execution			
Compute	K8s; Slurm	Kubernetes; Slurm; Ray	Gate on DT validation
Network	RESTCONF; NETCONF; gNMI	ONOS; ODL	Rollback if OSNR $>$ 1 dB
Power/cooling	OpenADR; BACnet	OpenADR VTN; EcoStruxure	Log all actions
Audit	OTEL; PROV-DM	Vector; Jaeger; OpenSearch	Store signed traces $\geq$ 12 mo

8 Sustainability-Aware Automation and Policy Implications

As AI infrastructure approaches environmental and resource limits, sustainability can no longer be treated as an external reporting obligation or a post-hoc optimization step. The preceding sections have shown that carbon intensity, water availability, and energy capacity function as hard feasibility constraints that determine whether AI operation is possible at a given site and time, not as efficiency metrics to be minimized within an otherwise unconstrained design space. This section examines the operational and policy implications of treating sustainability as a

first-class constraint embedded in the closed-loop control architecture developed in Section 7. It argues that sustainability-aware automation is not merely a technical enhancement but a governance mechanism that reshapes the relationship between infrastructure operators, regulators, and the physical environment, and that realizing its full potential requires policy frameworks designed for dynamic, telemetry-grounded enforcement rather than static, retrospective compliance.

8.1 Embedding sustainability in operational control

Sustainability-aware automation integrates carbon intensity, water usage, and energy availability into the same control loops that govern performance and reliability, ensuring that decisions reflect physical and environmental constraints in real time rather than against abstract annual targets [38, 57, 64, 74]. In this sense, sustainability-aware automation is not merely a technical enhancement but a governance mechanism that enables operational sovereignty over environmental constraints. Figure 7 makes this structure explicit by organizing the automation framework along a horizontal timescale axis. The figure should be read from top to bottom as a closed control loop: telemetry inputs at different characteristic timescales feed agentic control decisions, which produce validated operational actions, whose measured outcomes both update system state and provide compliance evidence to the policy layer. The figure reveals a constraint that is easy to state but frequently ignored in regulatory design: each sustainability signal operates at a characteristic timescale determined by the physical process it measures, and compliance must be assessed and enforced at that same timescale rather than collapsed to annual averages.

The four telemetry signals that feed the control system operate at distinct timescales, each determined by the physical process it represents. Grid carbon intensity — measured in grams of CO_2 equivalent per kilowatt-hour (gCO_2eq/kWh) — changes on sub-hourly timescales driven by generation dispatch transitions [17]. Commercial

providers such as WattTime and Electricity Maps expose these signals through APIs with update intervals of approximately five minutes; the architecture enforces a *freshness threshold* — a maximum allowable age for a telemetry reading before it is treated as stale and the dependent decision is held — of no more than five minutes for carbon intensity signals, reflecting the rate at which grid dispatch can shift a site across a regulatory carbon threshold. Energy headroom and power usage effectiveness (PUE) are sourced at hourly granularity through hardware management interfaces such as Redfish [22] and demand response platforms based on OpenADR 2.0b (Open Automated Demand Response) [38], a standard protocol that enables grid operators to signal power reduction requests to facility management systems (BMS, building management systems). Water permit utilization, measured in liters per megawatt-hour (L/MWh), is reported by facility BMS platforms using BACnet/IP (Building Automation and Control Networks over IP, ASHRAE Standard 135), a standard protocol for building system integration [75, 39]; this signal operates on daily-to-seasonal granularity because water permit ceilings are set and enforced seasonally by local regulators. Multi-year grid capacity and carbon budget signals are sourced from grid operator APIs and regulatory filings, and shape long-horizon planning decisions including site expansion feasibility and renewable energy procurement.

The agentic control system translates each telemetry signal into a corresponding control action at the appropriate timescale. At the sub-hourly level, carbon-aware scheduling shifts delay-tolerant batch analytics workloads to lower-carbon sites in real time; latency-sensitive workloads cannot be shifted because the propagation delay floor imposed by optical physics makes distant low-carbon sites physically infeasible for those workload classes [17], as analyzed in Section 4. Every proposed placement is validated by the digital twin before commitment to the southbound interface, ensuring that simulated carbon and performance consequences are confirmed before live execution. At the daily level, demand response and optical rerouting agents apply power capping, adjust cooling modes, and reroute optical traffic to lower-carbon sites using open control interfaces including OpenADR 2.0b for demand response and NETCONF [23] for optical network reconfiguration, with path feasibility validated through GNPy [67]. At the seasonal level, water-aware cooling control agents adjust cooling strategy and trigger pre-emptive workload

migration ahead of dry season periods informed by hydrometeorological forecasts [39, 75], using BACnet commands to building management systems. At the multi-year level, capacity and grid planning agents monitor the Feasible Sovereign Operating Region (FSOR) — the intersection of sovereignty, sustainability, and performance constraints within which the operator can deploy AI workloads — and evaluate renewable energy Power Purchase Agreements (PPAs), which are long-term contracts between electricity consumers and renewable generation owners that secure a guaranteed supply of low-carbon power at a fixed price [38, 74]. Long-horizon FSOR boundary monitoring uses digital twin simulation to project how tightening constraints will affect the feasible deployment region over planning horizons of three to ten years.

Executed operational actions reach physical infrastructure through validated southbound interfaces: workload placement decisions through the Kubernetes API and Ray [3], optical rerouting through NETCONF [23] and GNPy-validated paths, water permit throttling through BMS commands and workload migration triggers, and capacity expansion decisions through grid operator APIs and planning tools.

The policy and regulatory layer specifies compliance thresholds at the same timescale as the physical process each constraint governs. Sub-hourly carbon intensity ceilings are enforced per placement decision rather than against an annual average. Daily energy reporting obligations are met through PUE logs and demand response records consistent with the EU Code of Conduct on Data Center Energy Efficiency [25] and ISO 50001. Seasonal water permit ceilings are enforced daily and reviewed at the seasonal permit cycle. Multi-year carbon budget obligations align with net-zero pathways reported under Scope 1, Scope 2, and Scope 3 emissions accounting, as required under the EU Corporate Sustainability Reporting Directive (CSRD) [43].[1] [[1] Scope 1 covers direct emissions from owned

sources; Scope 2 covers indirect emissions from purchased energy; Scope 3 covers all other indirect emissions in the value chain, including embodied carbon in hardware and supply chain logistics.]

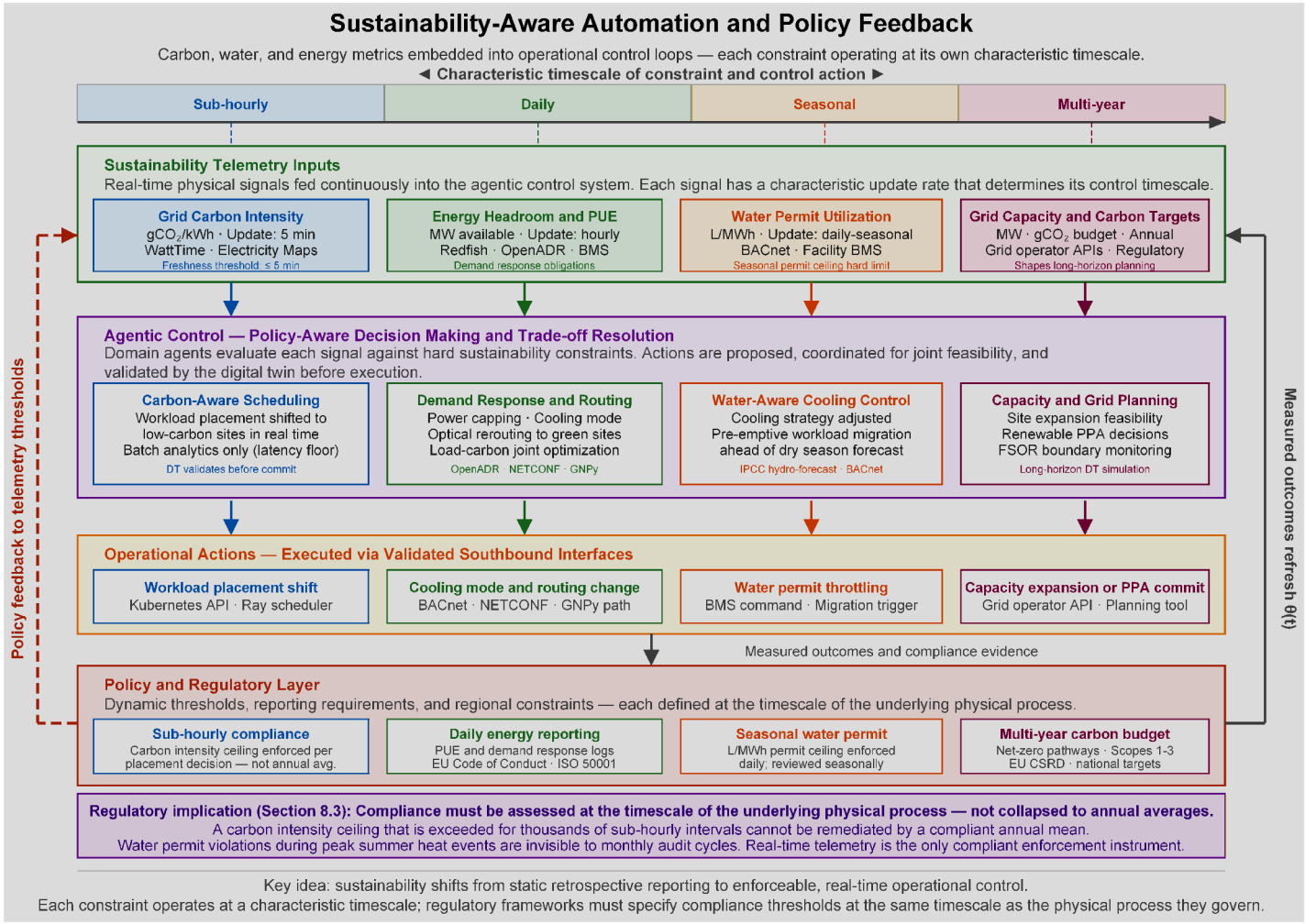


Figure 7: **Sustainability-aware automation and policy feedback, organized by characteristic timescale of constraint and control action.** The horizontal axis spans four timescale zones — sub-hourly, daily, seasonal, and multi-year — reflecting the characteristic update rate of each sustainability signal and the frequency at which corresponding control actions must operate. Row 1 shows the telemetry inputs feeding each zone: grid carbon

intensity (5-min updates) [17], energy headroom and PUE (hourly) [38], water permit utilization (daily-seasonal) [75, 39], and multi-year grid capacity and carbon budgets [74]. Row 2 shows the corresponding agentic control actions: carbon-aware workload scheduling, demand response and optical rerouting, water-aware cooling control with pre-emptive migration, and long-horizon capacity planning including FSOR boundary monitoring. Row 3 shows the executed operational actions through validated southbound interfaces. Row 4 shows the policy and regulatory layer, with compliance thresholds specified at the same timescale as the physical process they govern. Left and right feedback arrows close the loop: policy thresholds are fed back to update telemetry freshness requirements; measured outcomes refresh $\theta(t)$ in the observability layer. The purple box states the central regulatory implication: compliance must be assessed at the timescale of the physical process, not collapsed to annual averages.

The operational consequence of this embedding is a qualitative change in the nature of sustainability compliance. Under conventional reporting models, an operator measures aggregate carbon emissions over a quarter, compares them against a declared target, and adjusts strategy for the following period. Violations are identified retrospectively, after the environmental impact has already occurred, and remediation operates on timescales of months. Under sustainability-aware automation, the control system evaluates carbon intensity at each telemetry cycle and enforces the operator's carbon policy as a hard constraint on every placement and routing decision. A site whose grid intensity exceeds the policy threshold is removed from the feasible placement set immediately; a workload that would push water consumption above a permit limit is blocked before execution; a routing change that would increase carbon exposure above a declared budget triggers re-optimization rather than execution. Compliance is not reported after the fact but enforced within the control loop as an operational invariant: every placement, routing, and resource allocation decision must satisfy sustainability constraints at execution time, or it is rejected before reaching the infrastructure.

Specific control actions enabled by this embedding include temporal and spatial workload shifting in response to time-varying grid carbon intensity [57], dynamic adjustment of cooling strategies based on water availability and ambient wet-bulb temperature [75, 64], pre-emptive workload migration ahead of predicted water stress periods informed by hydrometeorological forecasts [39], and anticipatory reservation of routing capacity to low-carbon sites during periods when carbon intensity at current placement sites is forecast to rise. Each of these actions becomes available only when sustainability signals are treated as first-class control inputs on the same footing as latency and utilization metrics. When they are treated instead as reporting inputs consumed by a separate sustainability team, the control system lacks the information needed to act on them at the timescales that matter.

8.2 Unavoidable trade-offs and their governance

Embedding sustainability into operational control makes trade-offs explicit that were previously implicit and resolves them through explicit policy specification rather than through the accumulated effect of independent decisions made across organizational silos. These trade-offs define the boundary conditions of operational sovereignty: they determine which objectives can be jointly enforced by the control system and which require explicit policy relaxation. Three trade-offs are structurally unavoidable and warrant explicit treatment in any sustainability-aware control architecture.

The first is the *efficiency–resilience trade-off*. Minimizing carbon emissions drives workload concentration at sites with the cleanest grids, reducing the geographic diversity of active placements and increasing the exposure of the workload portfolio to local failures, grid instability events, or water stress periods at those sites. A pure carbon minimization objective without a resilience constraint will systematically underinvest in the redundancy needed to maintain service continuity under adverse conditions. The governance response is to encode a minimum diversity con-

straint alongside the carbon objective: the optimization must maintain feasible alternative placements for each workload class at sites in distinct failure domains, even if those alternatives have higher carbon intensity than the primary placement. The operator's policy specification determines how much carbon cost is acceptable to purchase a given level of resilience, making this trade-off explicit and auditable rather than implicit and opaque.

The second is the *carbon–latency trade-off*. The green-but-far effect analyzed in Section 4 means that the lowest-carbon sites are frequently not the closest to demand sources. For latency-sensitive workloads, this trade-off cannot be resolved by optimization: the propagation delay floor imposed by optical physics makes some low-carbon placements physically infeasible regardless of how the objective is weighted. For delay-tolerant workloads, the trade-off can be navigated through temporal shifting — running the workload at a time when a nearby site has lower carbon intensity — or through accepting a higher-latency but lower-carbon placement with explicit service-level objective relaxation. The policy specification must define which workload classes are subject to which trade-off resolution strategies, and the agentic system must enforce those definitions consistently across all placement decisions.

The third is the *autonomy–scale trade-off*. Prioritizing local operational sovereignty — restricting placements to domestically controlled sites, avoiding cross-border routing, and minimizing dependence on foreign optical infrastructure — reduces the feasible placement set and increases the probability of infeasibility under tight sustainability constraints. A region with a carbon-intensive grid and water stress cannot achieve both maximum sustainability compliance and maximum operational sovereignty simultaneously if its only compliant sites are abroad. This trade-off does not have a universal resolution; it reflects physical and geopolitical constraints that policy must navigate explicitly rather than dissolve through automation. The agentic control system's role is to make the trade-off visible

and quantifiable — specifying precisely which constraints cannot be satisfied within the sovereignty perimeter and what relaxation would restore feasibility — not to resolve it unilaterally [38, 74].

The three trade-offs described above are not independent optimization problems but competing pressures on a single bounded feasible region. Figure 8 visualizes this geometry. Each vertex of the triangle represents a primary objective — operational sovereignty, sustainability compliance, and AI performance — and the nested regions represent the space of jointly feasible configurations under progressively tighter constraints. The outermost region is the theoretical space in which no binding constraints are active. The intermediate region reflects the constrained feasible space imposed by regulatory carbon limits, water permit boundaries, and optical latency floors. The innermost region — the Feasible Sovereign Operating Region (FSOR) — is the intersection where all three objectives are simultaneously satisfiable under the operator's declared policy. The central point represents the optimal operating configuration within that region.

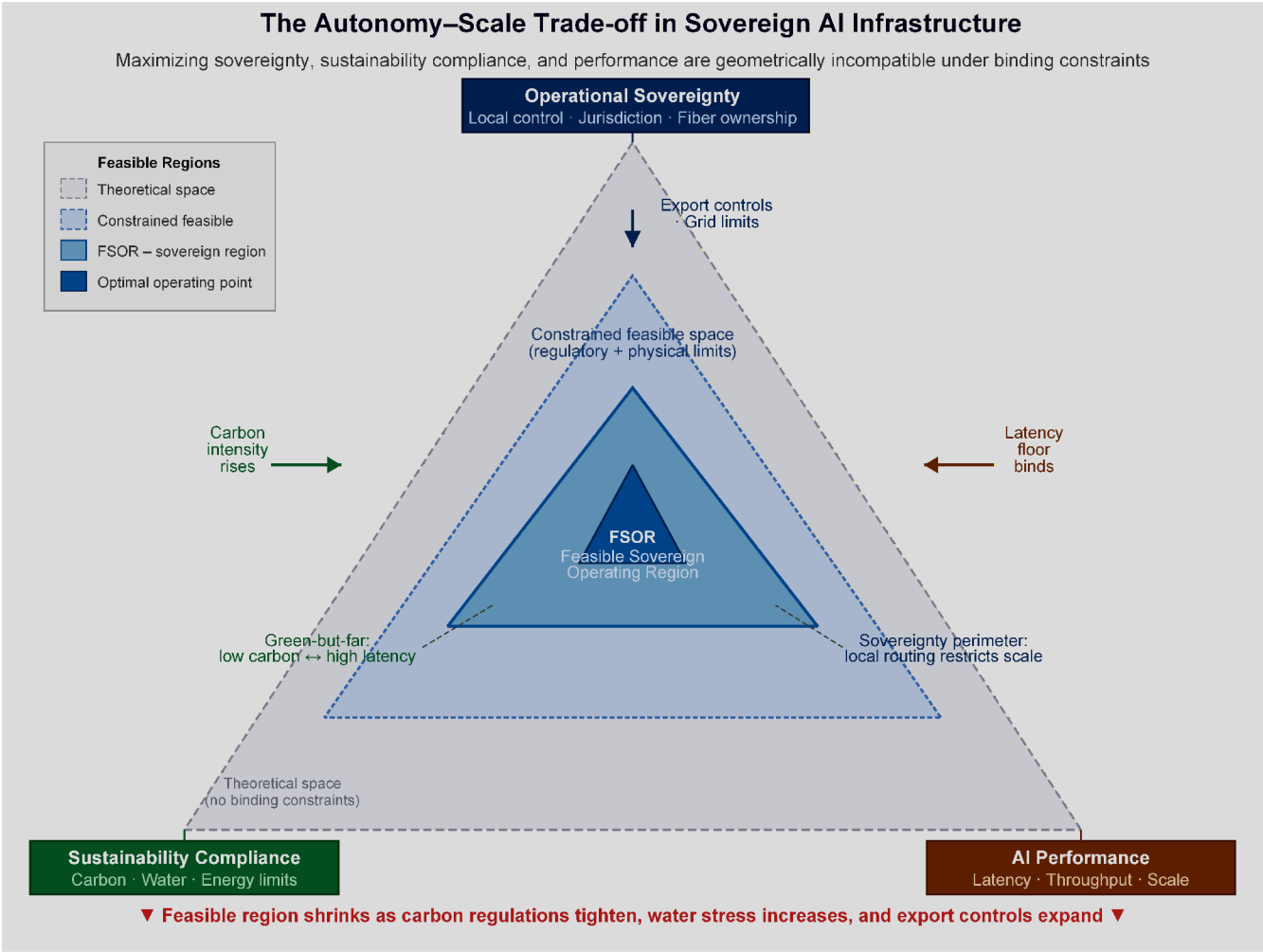


Figure 8: **The autonomy–scale trade-off in sovereign AI infrastructure.** Each vertex represents a primary objective: operational sovereignty (top), sustainability compliance (bottom left), and AI performance (bottom right). Nested regions show progressively tighter feasible spaces: the theoretical space with no binding constraints (outer, dashed grey), the constrained feasible space under regulatory and physical limits (middle, dashed blue), and the

Feasible Sovereign Operating Region — FSOR — where all constraints are jointly satisfied (inner, solid blue). The dark core marks the optimal operating point. Constraint pressure from rising carbon intensity, water stress, and export controls pushes inward from each edge, shrinking the FSOR. The green-but-far effect [38] bounds the sustainability–performance edge; sovereignty perimeter restrictions bound the sovereignty–performance edge. When all three constraints tighten simultaneously, the FSOR may contract to the empty set, requiring explicit policy relaxation rather than automated resolution.

As any single constraint tightens — through stricter carbon regulation, seasonal water stress, or expanded export controls on optical components — the FSOR shrinks. When constraints from different vertices tighten simultaneously, the region may contract to the empty set, at which point no jointly feasible configuration exists and the system must escalate to human authority for policy relaxation. The autonomy–scale trade-off is therefore not a design failure but a structural property of sovereign AI infrastructure: the geometry of competing constraints makes some combinations of objectives physically unreachable, and governance must specify which relaxations are acceptable rather than assume the trade-off away.

8.3 Implications for regulators

Sustainability-aware automation challenges the foundational assumptions of existing environmental compliance frameworks for digital infrastructure. Current regulatory models for data center sustainability are predominantly static and retrospective: they specify annual energy efficiency targets, require periodic carbon reporting against declared baselines, and assess compliance through audit processes that operate on timescales of months to years [43]. These mechanisms were designed for infrastructure whose operating parameters change slowly and whose

environmental impact can be meaningfully characterized by annual averages. Neither assumption holds for AI infrastructure operating under the sustainability-aware control architecture described in this tutorial.

Grid carbon intensity changes on sub-hourly timescales; a facility that reports a compliant annual average may have operated above its carbon threshold for thousands of intervals during high-demand periods when the grid mix was carbon-intensive. Water permit compliance measured monthly may conceal daily violations during heat waves. Energy efficiency metrics computed at annual granularity cannot capture the power dynamics of training workloads that fluctuate on much shorter timescales. Regulatory frameworks that rely on these coarse-grained measurements are not merely imprecise — they are structurally unable to detect the compliance failures that matter most in AI infrastructure environments.

A regulatory framework aligned with sustainability-aware automation would have four properties. First, it would recognize real-time telemetry as the authoritative source of compliance evidence, replacing retrospective self-reporting with continuous, machine-readable monitoring streams that regulators can access directly or through certified third-party auditors [25]. Second, it would specify compliance thresholds at the timescale of the underlying physical process: sub-hourly for carbon intensity, daily for water permit utilization, and instantaneous for power capacity constraints, rather than collapsing all dimensions to annual averages. Third, it would distinguish between hard regulatory limits — carbon intensity ceilings, water permit maxima — that must be enforced as instantaneous control constraints, and soft performance targets — annual average PUE, lifecycle carbon reduction trajectories — that can be evaluated at longer timescales without losing their regulatory meaning. Fourth, it would treat verifiable, auditable agentic control actions as compliance evidence: an operator who can demonstrate that their control system enforced a carbon intensity ceiling on every placement decision, with a complete action log and digital twin valida-

tion certificate for each, has provided stronger compliance evidence than one who reports an annual average that happens to fall below the threshold [57, 25].

Designing such a framework requires regulatory capacity that most jurisdictions currently lack: the technical expertise to evaluate telemetry pipeline integrity, the legal framework to recognize automated control actions as binding compliance instruments, and the institutional processes to audit agentic systems whose decision logic is distributed across multiple agents and a digital twin. Building this capacity is a prerequisite for enabling operational sovereignty over sustainability constraints at the timescales that matter, and it represents a significant governance investment that should begin before AI infrastructure scale makes it urgent.

8.4 Implications for infrastructure operators

For infrastructure operators, sustainability-aware automation transforms sustainability from a compliance function into an operational capability with direct consequences for cost, reliability, and competitive position [38, 64]. In this context, sustainability becomes a dimension of operational sovereignty: the ability to enforce environmental constraints through control decisions rather than through retrospective reporting. The transition requires investment across three dimensions that are interdependent and must be developed in coordination.

The first dimension is *telemetry infrastructure*. Operators who lack real-time carbon intensity signals, water permit monitoring, or sub-minute power draw measurements cannot embed sustainability into control loops regardless of how sophisticated their optimization and agentic systems are. The telemetry gaps identified in Section 5 — missing freshness certification, absence of cross-domain timestamp alignment, vendor-specific schemas that resist normal-

ization — are not merely monitoring deficiencies; they are control limitations that reduce the precision and reliability of carbon and water constraint enforcement [20].

The second dimension is *control architecture*. Operators whose sustainability metrics flow into a reporting dashboard rather than into the objective function and constraint set of their placement optimizer have not embedded sustainability into control; they have added a sustainability display to an unchanged control system. Genuine embedding requires that carbon intensity and water permit utilization appear in the same mathematical structure as latency and power capacity constraints, as hard limits that eliminate infeasible configurations rather than as soft penalties that can be overridden by performance pressure. Achieving this requires either redesigning existing orchestration systems or deploying a sustainability-aware control layer above them that intercepts and filters placement decisions before execution.

The third dimension is *organizational process*. Sustainability-aware automation produces decisions at timescales and volumes that exceed human review capacity: a system managing hundreds of sites and thousands of workloads may make millions of sustainability-relevant placement decisions per day. Operators must develop governance processes that define the scope of automated authority, specify escalation paths for constraint conflicts and infeasibility events, establish audit procedures for reviewing agentic decision logs, and maintain the policy specifications that bound agent behavior. These processes are as important as the technical architecture; an agentic system operating without clear policy specifications and governance oversight is not an instrument of sovereign control but a source of unaccountable automation.

8.5 Implications for hyperscalers

Hyperscalers occupy a structurally distinct position in the sustainability governance landscape. Their scale enables capital investment in advanced telemetry, optimization, and digital twin infrastructure that is inaccessible to smaller operators, and their geographic footprint across multiple grid regions enables sophisticated sustainability arbitrage — shifting workloads in real time to the lowest-carbon available sites across a global infrastructure portfolio [57, 74]. At the same time, their scale amplifies environmental impact: a marginal efficiency improvement in a hyperscaler's carbon-aware scheduler affects more aggregate emissions than a comparable improvement in any individual regional operator's system. Hyperscalers therefore carry disproportionate responsibility for demonstrating that sustainability-aware automation works at scale and for contributing the operational data that can inform the regulatory frameworks discussed above.

The geopolitical dimension of hyperscaler sustainability practice is underappreciated. When a hyperscaler's carbon-aware scheduler shifts workloads away from a high-carbon region, it reduces that region's AI compute revenue and reinforces the competitive disadvantage of carbon-intensive grids. When it shifts workloads toward low-carbon regions, it increases those regions' effective AI capacity and strengthens their position in the infrastructure sovereignty landscape. These allocation effects are not neutral: they reflect and reinforce the structural inequality between regions with clean energy access and regions without it, examined in Section 3. Regulators in carbon-intensive regions may find that hyperscaler carbon-aware scheduling exacerbates their AI infrastructure disadvantage, creating incentives to regulate or restrict sustainability-driven workload shifting in ways that conflict with global emissions reduction objectives.

This tension between national AI sovereignty and global sustainability optimization reflects a mismatch between geographically bounded governance and globally optimized control systems, and it has no clean technical resolu-

tion. It therefore requires explicit policy negotiation at the international level [6, 38]. Such negotiation is likely to center on mechanisms that align global efficiency with national sovereignty objectives, including carbon accounting and credit frameworks that recognize cross-border workload shifting, minimum domestic capacity requirements for critical AI functions, and technology transfer or infrastructure co-investment obligations tied to hyperscaler deployment. These instruments do not eliminate the tension between sovereignty and sustainability, but they define how it can be governed in practice, enabling both objectives to become operational rather than mutually exclusive.

8.6 Sustainability, sovereignty, and a forward policy agenda

The analysis in this tutorial converges on a conclusion that connects the sustainability and sovereignty dimensions of AI infrastructure in a way that neither literature has fully articulated. Sustainability constraints are sovereignty constraints. A region whose grid is carbon-intensive, whose water resources are stressed, or whose energy infrastructure cannot support AI-scale power demand faces sovereignty limits that are physical rather than legal or political in nature. No regulatory framework, no data localization requirement, and no model ownership policy can substitute for the physical capacity to power, cool, and connect AI infrastructure within locally defined environmental limits. Conversely, a region that achieves effective sustainability-aware control of its AI infrastructure — enforcing carbon and water limits in real time, shifting workloads in response to grid signals, and maintaining digital twin models that make infrastructure behavior verifiable — demonstrates operational sovereignty: control authority grounded in the ability to observe, decide, validate, and act within physical constraints rather than in formal ownership alone.

This convergence implies a forward policy agenda organized around four priorities. First, *grid decarbonization as AI sovereignty policy*: investments in renewable energy capacity, transmission infrastructure, and grid flexibility are

not merely environmental measures but strategic enablers of sovereign AI deployment that expand the Feasible Sovereign Operating Region (FSOR) for regions currently constrained by carbon-intensive generation. Second, *telemetry standardization as regulatory infrastructure*: open, interoperable telemetry standards for data center power, cooling, water, and network state are prerequisites for both sustainability-aware automation and dynamic regulatory compliance; their development should be treated as public infrastructure rather than left solely to vendor-driven standardization processes. Third, *dynamic compliance frameworks*: regulatory agencies should develop the technical and legal capacity to recognize real-time telemetry and auditable agentic control actions as compliance evidence, replacing annual reporting cycles with continuous, machine-readable compliance streams that reflect actual operational behavior. Fourth, *sovereignty-aware sustainability policy*: international negotiations on AI infrastructure sustainability should account for the differential impact of global carbon optimization on regional AI sovereignty, developing mechanisms — such as clean energy investment obligations, technology transfer provisions, or carbon credit frameworks tied to AI infrastructure deployment — that allow sustainability objectives to be pursued without systematically disadvantaging regions whose grid decarbonization is incomplete.

Together, these priorities define a governance agenda in which sustainability and sovereignty are pursued as mutually reinforcing rather than competing objectives, grounded in the physical realities of AI infrastructure and implemented through the telemetry-driven, agentic control architecture described in this tutorial. In this framework, sovereignty is not asserted through ownership alone but realized through continuous, validated control of infrastructure behavior under constraint.

9 Conclusion

The rapid expansion of artificial intelligence has shifted the meaning of sovereignty from a question of data and algorithms to a question of infrastructure. As AI systems scale, their feasibility and impact are increasingly determined by access to power, cooling, water, and optical connectivity, together with the operational capability to manage these resources under environmental and geopolitical constraints. AI sovereignty is therefore not an abstract policy objective, but a property that emerges from the physical and operational structure of infrastructure systems.

This tutorial has presented a cross-layer view of AI sovereignty grounded in engineering practice. It has shown that AI-oriented data centers, optical transport networks, and energy systems form a tightly coupled substrate whose constraints must be satisfied simultaneously. Sustainability limits, particularly those related to energy, carbon intensity, and water usage, act as hard boundaries that define where AI can be deployed. Optical networks, in turn, determine how far and how reliably these capabilities can be extended across regions. Sovereignty thus emerges from the joint feasibility of these layers, rather than from any single dimension in isolation. The Feasible Sovereign Operating Region (FSOR) formalizes this emergence: it is the set of workload configurations a given infrastructure can actually sustain under its physical, environmental, and regulatory endowment, computable from real-time telemetry and directly actionable as a planning instrument.

A central contribution of this work is to position telemetry and agentic AI as the operational foundation of sovereign infrastructure. Telemetry transforms infrastructure into a continuously observable system, while agent-based control and digital twins enable coordinated and validated action across compute, network, and energy domains. This

closed-loop capability allows operators to enforce constraints, manage trade-offs, and adapt to changing conditions in real time, making sovereignty a function of visibility, decision authority, and execution capability.

These insights have direct implications for both engineering and policy. For the engineering community, AI system design must move beyond siloed optimization toward integrated, cross-layer architectures in which performance, efficiency, sustainability, and control are jointly considered. Real-time observability and closed-loop control are no longer optional features, but fundamental design requirements for operating AI systems at scale within physical limits. The green-but-far effect — the structural tension between sustainability eligibility and latency admissibility that makes joint placement–routing optimization a necessity rather than a convenience — illustrates why cross-layer integration cannot be deferred to post-design optimization.

For policymakers and regulators, the results highlight the need to align governance frameworks with operational reality. Static notions of compliance based on data location or ownership are insufficient in systems constrained by dynamic physical conditions. Effective sovereignty requires mechanisms that enable verifiable, real-time control over infrastructure behavior, allowing environmental targets, reliability requirements, and autonomy objectives to be enforced in practice rather than declared in principle.

More broadly, this work reframes AI infrastructure as a feasibility-constrained system in which physical limits, environmental boundaries, and control capabilities jointly determine what can be built and operated. In this context, sovereignty is not defined by isolation, but by the ability to exercise effective control within these constraints. The future of AI will therefore be shaped as much by power systems, cooling technologies, and optical networks as by

models and algorithms. Understanding, designing, and governing these foundations is essential for building AI ecosystems that are sustainable, resilient, and operationally sovereign.

References

[1] AddOn Networks (2024)
Data sovereignty and fiber optics.
Note: https://www.addonnetworks.com/solutions/insights/data-sovereignty-fiber-optics-article (https://www.addonnetworks.com/solutions/insights/data-sovereignty-fiber-optics-article)Accessed 2026
Cited by: §4.4.

[2] E. Androulaki, A. Barger, V. Bortnikov, C. Cachin, K. Christidis, A. De Caro, D. Enyeart, C. Ferris, G. Laventman, Y. Manevich, S. Muralidharan, C. Murthy, B. Nguyen, M. Sethi, G. Singh, K. Smith, A. Sorniotti, C. Stathakopoulou, M. Vukolić, S. W. Cocco, and J. Yellick (2018)
Hyperledger fabric: a distributed operating system for permissioned blockchains.
In Proceedings of the Thirteenth EuroSys Conference,
pp. 1–15.
External Links: Link (http://dx.doi.org/10.1145/3190508.3190538), Document (https://dx.doi.org/10.1145/3190508.3190538)
Cited by: §7.5.

[3] Anyscale (2024)
Ray: A Unified Framework for Scaling AI and Python Applications.
External Links: Link (https://www.ray.io)
Cited by: §8.1.

[4] C. Arora, M. Sorel, and P. Sachdeva (2025)
The next big shifts in AI workloads and hyperscaler strategies.
Note: https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-next-big-shifts-in-ai-workloads-and-hyperscaler-strategies (https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-next-big-shifts-in-ai-workloads-and-hyperscaler-strategies)
Cited by: §2.1.

[5] ASHRAE (2020)
ANSI/ASHRAE Standard 135-2020: A Data Communication Protocol for Building Automation and Control Networks (BACnet).
Technical report
American Society of Heating, Refrigerating and Air-Conditioning Engineers.
External Links: Link (https://www.ashrae.org/technical-resources/%0Abookstore/bacnet)
Cited by: §5.2.

[6] M. S. Bakare, A. Abdulkarim, A. N. Shuaibu, and M. M. Muhamad (2024)
Energy management controllers: strategies, coordination, and applications.
Energy Informatics **7** (1), pp. 57.
External Links: Document (https://dx.doi.org/10.1186/s42162-024-00357-9), ISSN 2520-8942
Cited by: §8.5.

[7] M. Behringer, M. Pritikin, S. Bjarnason, A. Clemm, B. Carpenter, and S. Jiang (2015)
Autonomic networking: definitions and design goals.

Internet Research Task Force (IRTF).
Note: RFC 7575
External Links: Document (https://dx.doi.org/10.17487/RFC7575), Link (https://datatracker.ietf.org/doc/html/rfc7575)
Cited by: §1.3, §5, §6.

[8] R. E. Bixby (2012)
A brief history of linear and mixed-integer programming computation.
In Documenta Mathematica Extra Volume ISMP (2012),
pp. 107–121.
External Links: Link (https://ems.press/content/book-chapter-files/27357), Document (https://dx.doi.org/10.4171/DM/2012)
Cited by: §6.2.

[9] R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, M. S. Bernstein, J. Bohg, A. Bosselut, E. Brunskill, E. Brynjolfsson, S. Buch, D. Card, R. Castellon, N. Chatterji, A. Chen, K. Creel, J. Q. Davis, D. Demszky, C. Donahue, M. Doumbouya, E. Durmus, S. Ermon, J. Etchemendy, K. Ethayarajh, L. Fei-Fei, C. Finn, T. Gale, L. Gillespie, K. Goel, N. Goodman, S. Grossman, N. Guha, T. Hashimoto, P. Henderson, J. Hewitt, D. E. Ho, J. Hong, K. Hsu, J. Huang, T. Icard, S. Jain, D. Jurafsky, P. Kalluri, S. Karamcheti, G. Keeling, F. Khani, O. Khattab, P. W. Koh, M. Krass, R. Krishna, R. Kuditipudi, A. Kumar, F. Ladhak, M. Lee, T. Lee, J. Leskovec, I. Levent, X. L. Li, X. Li, T. Ma, A. Malik, C. D. Manning, S. Mirchandani, E. Mitchell, Z. Munyikwa, S. Nair, A. Narayan, D. Narayanan, B. Newman, A. Nie, J. C. Niebles, H. Nilforoshan, J. Nyarko, G. Ogut, L. Orr, I. Papadimitriou, J. S. Park, C. Piech, E. Portelance, C. Potts, A. Raghunathan, R. Reich, H. Ren, F. Rong, Y. Roohani, C. Ruiz, J. Ryan, C. Ré, D. Sadigh, S. Sagawa, K. Santhanam, A. Shih, K. Srinivasan, A. Tamkin, R. Taori, A. W. Thomas, F. Tramèr, R. E. Wang, W. Wang, B. Wu, J. Wu, Y. Wu, S. M. Xie, M. Yasunaga, J. You, M. Zaharia, M. Zhang, T. Zhang, X. Zhang, Y. Zhang, L. Zheng, K. Zhou, and P. Liang (2022)
On the opportunities and risks of foundation models.

External Links: 2108.07258, Link (https://arxiv.org/abs/2108.07258)
Cited by: §2.1.

[10] R. Buyya, A. Beloglazov, and J. Abawajy (2013)
Energy-efficient management of data center resources for cloud computing.
IEEE Transactions on Parallel and Distributed Systems **24** (7), pp. 1396–1407.
Cited by: §3.1.

[11] J. Case, M. Fedor, M. Schoffstall, and J. Davin (1990)
Simple network management protocol (snmp).
Request for Comments, IETF.
Note: https://datatracker.ietf.org/doc/html/rfc1157 (https://datatracker.ietf.org/doc/html/rfc1157)[STANDARDS-TRACK]
Cited by: §5.2.

[12] X. Chen, X. Wang, A. Colacelli, M. Lee, and L. Xie (2025)
Electricity demand and grid impacts of ai data centers: challenges and prospects.
arXiv preprint arXiv:2509.07218.
Cited by: §1.2, §1.2, §1, §2.2, §2.2, §3.2.

[13] S. B. Chetty, D. Grace, S. Saunders, P. Harris, E. E. Tsiropoulou, T. Quek, and H. Ahmadi (2025)
Sovereign AI for 6g: towards the future of AI-native networks.
External Links: 2509.06700, Document (https://dx.doi.org/10.48550/arXiv.2509.06700), Link (https://arxiv.org/abs/2509.06700)
Cited by: §4.1, §4.2, §4.2, §4.5.

[14] E. Choukse, B. Warrier, S. Heath, L. Belmont, A. Zhao, H. A. Khan, B. Harry, M. Kappel, R. J. Hewett, K. Datta, Y. Pei, C. Lichtenberger, J. Siegler, D. Lukofsky, Z. Kahn, G. Sahota, A. Sullivan, C. Frederick, H. Thai, R. Naughton, D. Jurnove, J. Harp, R. Carper, N. Mahalingam, S. Varkala, A. G. Kumbhare, S.

Desai, V. Ramamurthy, P. Gottumukkala, G. Bhatia, K. Wildstone, L. Olariu, I. Incorvaia, A. Wetmore, P. Ram, M. Raghuraman, M. Ayna, M. Kendrick, R. Bianchini, A. Hurst, R. Zamani, X. Li, M. Petrov, G. Oden, R. Carmichael, T. Li, A. Gupta, P. Patel, N. Dattani, L. Marwong, R. Nertney, H. Kobayashi, J. Liott, M. Enev, D. Ramakrishnan, I. Buck, and J. Alben (2025)
Power stabilization for AI training datacenters.
Note: https://arxiv.org/abs/2508.14318 (https://arxiv.org/abs/2508.14318)arXiv:2508.14318
External Links: 2508.14318
Cited by: §2.2, §3.2, §5.2, §5.3.

[15] Cloud Security Web (2025)
Understanding telemetry for cloud infrastructure optimization.
Note: https://cloudsecurityweb.com/articles/2025/04/13/understanding-telemetry-for-cloud-infrastructure-optimization/ (https://cloudsecurityweb.com/articles/2025/04/13/understanding-telemetry-for-cloud-infrastructure-optimization/)Online technical article
Cited by: §5.5.

[16] Cornell Chronicle (2025)
Roadmap shows the environmental impact of ai data center boom.
Note: Accessed: 2026-01-30
External Links: Link (https://news.cornell.edu/stories/2025/11/roadmap-shows-environmental-impact-ai-data-center-boom)
Cited by: §3.3.

[17] L. Cote and A. Sun (2025)
Locational marginal emissions for carbon-aware data center operations in large-scale power grids.
External Links: 2512.18819, Link (https://arxiv.org/abs/2512.18819)
Cited by: §3.3, §5.2, §6.1, §7.3, §7.7, Figure 7, §8.1, §8.1.

[18] S. Cruzes (2024)
Failure management overview in optical networks.
IEEE Access **12**, pp. 169170–169193.
External Links: Document (https://dx.doi.org/10.1109/ACCESS.2024.3498704)
Cited by: §4.5.

[19] S. Cruzes (2025)
Revolutionizing optical networks: the integration and impact of large language models.
Optical Switching and Networking **57**, pp. 100812.
External Links: ISSN 1573-4277, Document (https://dx.doi.org/https%3A//doi.org/10.1016/j.osn.2025.100812), Link (https://www.sciencedirect.com/science/article/pii/S1573427725000190)
Cited by: §1.3, §6.1, §6.3, §6.3.

[20] S. Cruzes (2026)
Telemetry and agentic AI: foundations for optical network automation.
IEEE Access **14**, pp. 8800–8838.
External Links: Document (https://dx.doi.org/10.1109/ACCESS.2025.3649768)
Cited by: §1.3, §4.1, §5.1, §5.2, §5.2, §5.3, §5, §6.1, §6.1, §6.2, §6.2, §6.3, §6, §7.4, §8.4.

[21] DataCenters.com (2026)
Inside the AI compute stack: why infrastructure not models is the real bottleneck.
Note: https://www.datacenters.com/news/inside-the-ai-compute-stack-why-infrastructure-not-models-is-the-real-bottleneck (https://www.datacenters.com/news/inside-the-ai-compute-stack-why-infrastructure-not-models-is-the-real-bottleneck)Online news article
Cited by: §1.2, §4.5.

[22] DMTF (2024)
Redfish specification.

Standard Specification
Technical Report DSP0266, Distributed Management Task Force.
External Links: Link (https://www.dmtf.org/sites/default/files/standards/documents/DSP0266_1.21.1.html)
Cited by: §5.2, §6.1, §7.3, §7.7, §8.1.

[23] R. Enns, M. Björklund, J. Schönwälder, and A. Bierman (2011)
Network Configuration Protocol (NETCONF).
RFC
Technical Report 6241, IETF.
External Links: Document (https://dx.doi.org/10.17487/RFC6241), Link (https://datatracker.ietf.org/doc/html/rfc6241)
Cited by: §5.2, §8.1, §8.1.

[24] M. Esposito (2025)
AI geopolitics and data in the era of technological rivalry.
Note: https://www.weforum.org/stories/2025/07/ai-geopolitics-data-centres-technological-rivalry/ (https://www.weforum.org/stories/2025/07/ai-geopolitics-data-centres-technological-rivalry/)World Economic Forum online article
Cited by: §1.2.

[25] European Commission, Joint Research Centre (2025)
2025 Best Practice Guidelines for the EU Code of Conduct on Data Centre Energy Efficiency.
Technical report
Technical Report JRC141521, Publications Office of the European Union.
Note: Version 16.1.0, 1 January 2025
External Links: Document (https://dx.doi.org/10.2760/9449356), Link (https://publications.jrc.ec.europa.eu/repository/handle/JRC141521)
Cited by: §8.1, §8.3.

[26] European Parliament and Council of the European Union (2016)

Regulation (EU) 2016/679 of the european parliament and of the council on the protection of natural persons with regard to the processing of personal data (general data protection regulation).
Technical report
Technical Report L 119, Official Journal of the European Union.
Note: OJ L 119, 4.5.2016, pp. 1–88
External Links: Link (https://eur-lex.europa.eu/eli/reg/2016/679/oj)
Cited by: §1.1.

[27] P. Fraga-Lamas and T. M. Fernández-Caramés (2019)
A review on blockchain technologies for an advanced and cyber-resilient automotive industry.
IEEE Access **7** (), pp. 17578–17598.
External Links: Document (https://dx.doi.org/10.1109/ACCESS.2019.2895302)
Cited by: §7.5, §7.5.

[28] S. Fratini and O. Hine (2024)
Digital sovereignty: a descriptive analysis and a critical evaluation of existing models.
Digital Society.
External Links: Document (https://dx.doi.org/10.1007/s44206-024-00146-7), Link (https://doi.org/10.1007/s44206-024-00146-7)
Cited by: Figure 1, §1.1, §1.1.

[29] FutureCIO (2025)
Impact of AI workload scalability and sovereignty on data centre operations.
Note: https://futurecio.tech/impact-of-ai-workload-scalability-and-sovereignty-on-data-centre-operations/ (https://futurecio.tech/impact-of-ai-workload-scalability-and-sovereignty-on-data-centre-operations/)Online article
Cited by: §1.1.

[30] J. Ghez and O. Chatain (2024)

AI and sovereignty: the geopolitical power of submarine cables.
HEC Paris.
Note: https://www.hec.edu/en/dare/geopolitics-public-affairs/ai-and-sovereignty-geopolitical-power-submarine-cables (https://www.hec.edu/en/dare/geopolitics-public-affairs/ai-and-sovereignty-geopolitical-power-submarine-cables)Research article
Cited by: §4.4.

[31] Green Software Foundation (2021)
Carbon-aware SDK.
Note: https://github.com/Green-Software-Foundation/carbon-aware-sdk (https://github.com/Green-Software-Foundation/carbon-aware-sdk)Accessed: 2026-03-29
Cited by: §5.2, §6.1, §7.7.

[32] R. Gu, Z. Yang, and Y. Ji (2020)
Machine learning for intelligent optical networks: a comprehensive survey.
Journal of Network and Computer Applications **157**, pp. 102576.
External Links: Document (https://dx.doi.org/10.1016/j.jnca.2020.102576), ISSN 1084-8045
Cited by: §4.1, §5.1.

[33] J. Hoxha, M. Thanasi-Boçe, and T. Khalifa (2025)
A deployment-aware framework for carbon- and water-efficient llm serving.
Sustainability **17** (23), pp. 10473.
Cited by: §1.2, §1, §2.1, §4.5.

[34] IBM (2026)
Introducing IBM sovereign core: a new software foundation for sovereignty.
Note: https://www.ibm.com/new/announcements/introducing-ibm-sovereign-core-a-new-software-foundation-for-sovereignty (https://www.ibm.com/new/announcements/introducing-ibm-sovereign-core-a-new-software-f

oundation-for-sovereignty)Product announcement
Cited by: §1.2.

[35] IEEE Communications Society (2025)
Sovereign ai infrastructure for telecom companies: implementation and challenges.
Note: https://techblog.comsoc.org/2025/12/17/sovereign-ai-infrastructure-for-telecom-companies-implementation-and-challenges/ (https://techblog.comsoc.org/2025/12/17/sovereign-ai-infrastructure-for-telecom-companies-implementation-and-challenges/)Accessed 2026
Cited by: §4.4.

[36] IEEE Spectrum (2026)
RF over fiber.
IEEE Spectrum.
External Links: Link (https://spectrum.ieee.org/rf-over-fiber)
Cited by: §4.2, §4.2.

[37] Intel Corporation, Hewlett-Packard, NEC, and Dell (2013)
Intelligent platform management interface specification: second generation.
Technical report
Technical Report v2.0, Document Revision 1.1, Intel Corporation.
External Links: Link (https://www.intel.com.tw/content/dam/www/public/us/en/documents/product-briefs/ipmi-second-gen-interface-spec-v2-rev1-1.pdf)
Cited by: §5.2.

[38] International Energy Agency (IEA) (2024)
Data centres and data transmission networks.
Note: Accessed: 2026-01-30
External Links: Link (https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks)

Cited by: §1.2, §1, §2.2, §2.2, §2.2, §2.4, §3.1, §3.1, §3.3, §3.4, §3.5, §7.2, Figure 7, Figure 8, §8.1, §8.1, §8.1, §8.2, §8.4, §8.5.

[39] IPCC (2021)
Climate change 2021: the physical science basis. contribution of working group I to the sixth assessment report of the intergovernmental panel on climate change.
Technical report
Cambridge University Press.
External Links: Document (https://dx.doi.org/10.1017/9781009157896)
Cited by: §3.1, §3.4, Figure 7, §8.1, §8.1, §8.1.

[40] M. Kiasari and H. Aly (2026)
Agentic artificial intelligence for smart grids: a comprehensive review of autonomous, safe, and explainable control frameworks.
Energies **19** (3), pp. 617.
External Links: Document (https://dx.doi.org/10.3390/en19030617), Link (https://doi.org/10.3390/en19030617)
Cited by: §1.3, §6.2, §6.5.

[41] A. M. Kirubakaran, A. Parthasarathy, N. Saksena, R. S. Bodala, A. Deshpande, S. Malempati, S. K. R. Carimireddy, and A. Mazumder (2025)
Governing cloud data pipelines with agentic AI.
Eighth Sense Research Group.
Note: Zenodo preprint
External Links: Document (https://dx.doi.org/10.5281/zenodo.18048728), Link (https://zenodo.org/doi/10.5281/zenodo.18048728)
Cited by: §6.5.

[42] G. Korde, T. Konar, and J. Lamba (2025)

Sovereign data center strategy.
Nokia Bell Labs Consulting.
Note: White paper, Document code: CID215164
External Links: Link (https://www.nokia.com/asset/i/215164/)
Cited by: §1.2, §1.2, §4.1, §4, §7.2.

[43] K. Mayegun and C. Nwanevu (2025)
Harnessing big data and AI to revolutionize sustainability accounting and integrated corporate financial reporting.
International Journal of Computer Applications Technology and Research **14** (6), pp. 111–128.
External Links: Document (https://dx.doi.org/10.7753/IJCATR1406.1008)
Cited by: §8.1, §8.3.

[44] MIT Technology Review (2026)
Everyone wants AI sovereignty, no one can truly have it.
MIT Technology Review.
Note: Accessed: Jan. 2026
External Links: Link (https://www.technologyreview.com/2026/01/21/1131513/everyone-wants-ai-sovereignty-no-one-can-truly-have-it/)
Cited by: Figure 1, §1.2.

[45] A. Mitrovska, B. Shariati, A. Jafari, P. Safari, J. K. Fischer, and R. Freund (2025)
Network data sharing: a governance framework for ensuring data sovereignty and privacy compliance.
Journal of Optical Communications and Networking **17** (11), pp. 1019–1031.
External Links: Document (https://dx.doi.org/10.1364/JOCN.559523)
Cited by: §5.4.

[46] Modbus Organization (2006)

MODBUS application protocol specification.
Technical report
Technical Report V1.1b, Modbus Organization.
External Links: Link (https://www.modbus.org/file/secure/modbusprotocolspecification.pdf)
Cited by: §5.2.

[47] L. Moreau, P. Missier, K. Belhajjame, R. B'Far, J. Cheney, S. Coppens, S. Cresswell, Y. Gil, P. Groth, G. Klyne, T. Lebo, J. McCusker, D. McGuinness, J. Myers, S. Sahoo, and C. Tilmes (2013)
PROV-DM: The PROV Data Model.
W3C Recommendation
World Wide Web Consortium (W3C).
Note: W3C Recommendation 30 April 2013
External Links: Link (https://www.w3.org/TR/prov-dm/)
Cited by: §7.5.

[48] Naddod (2026)
Understanding scale-up vs. scale-out in AI infrastructure.
Naddod.
External Links: Link (https://www.naddod.com/blog/scale-up-vs-scale-out-in-ai-infrastructure)
Cited by: §4.2.

[49] P. Nalage (2025)
Agentic digital twins: self-evolving models for autonomous systems.
Well Testing Journal **34** (S3), pp. 227–244.
External Links: Link (https://welltestingjournal.com/index.php/WT/article/view/183)
Cited by: §6.5.

[50] OpenConfig Working Group (2016)

OpenConfig: Vendor-Neutral, Model-Driven Network Management.
Note: Open-source data modeling initiative for network device configuration and telemetry. Available at: https://www.openconfig.net (https://www.openconfig.net)
External Links: Link (https://www.openconfig.net)
Cited by: §5.2.

[51] OpenTelemetry (2025)
OpenTelemetry specification.
Note: https://opentelemetry.io/docs/specs/otel/ (https://opentelemetry.io/docs/specs/otel/)Accessed: 2026-03-29
Cited by: §6.1, §7.7.

[52] J. O'Donnell and C. Crownhart (2025)
We did the math on AI's energy footprint. Here's the story you haven't heard.
Note: https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/ (https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/)
Cited by: §1.2.

[53] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean (2021)
Carbon emissions and large neural network training.
arXiv preprint arXiv:2104.10350.
Cited by: §1.2, §1, §2.1.

[54] Photonics21 (2026)
Investing in Light: the vision to secure europe's competitiveness in the global tech race.
Technical report
Photonics21.

External Links: Link (https://www.photonics21.org/download/ppp-services/photonics-downloads/Photonics21_Position_Paper_Investing_in_Light_The_Vision_to_Secure_Europes_Competitiveness_in_the_Global_Tech_Race.pdf)
Cited by: §7.2.

[55] Point2 Technology (2026)
Futuring interconnect infrastructure for AI: RF transmission over plastic cable surpasses copper and optics at terabit scale.
Signal Integrity Journal.
External Links: Link (https://www.signalintegrityjournal.com/articles/4011-futuring-interconnect-infrastructure-for-ai-rf-transmission-over-plastic-cable-surpasses-copper-and-optics-at-terabit-scale)
Cited by: §4.2.

[56] T. I. Project (2020)
Technical specification tr-547 – tapi v2.1.3 reference implementation agreement.
Technical report
Open Networking Foundation.
External Links: Link (https://opennetworking.org/wp-content/uploads/2020/08/TR-547-TAPI-v2.1.3-Reference-Implementation-Agreement-1.pdf)
Cited by: §7.7.

[57] A. Radovanovic, R. Koningstein, I. Schneider, B. Chen, A. Duarte, B. Roy, D. Xiao, M. Haridasan, P. Hung, N. Care, S. Talukdar, E. Mullen, K. Smith, M. Cottman, and W. Cirne (2021)
Carbon-aware computing for datacenters.
Note: https://arxiv.org/abs/2106.11750 (https://arxiv.org/abs/2106.11750)arXiv:2106.11750
External Links: 2106.11750
Cited by: §8.1, §8.1, §8.3, §8.5.

[58] T. P. Raptis and A. Passarella (2023)

A survey on networked data streaming with apache kafka.
IEEE Access **11** (), pp. 85333–85350.
External Links: Document (https://dx.doi.org/10.1109/ACCESS.2023.3303810)
Cited by: §6.1, §7.7.

[59] E. B. Sanjuan, I. A. Cardiel, J. A. Cerrada, and C. Cerrada (2020)
Message queuing telemetry transport (mqtt) security: a cryptographic smart card approach.
IEEE Access **8** (), pp. 115051–115062.
External Links: Document (https://dx.doi.org/10.1109/ACCESS.2020.3003998)
Cited by: §7.7.

[60] R. Shakir, A. Shaikh, P. Borman, M. Hines, C. Lebsack, and C. Morrow (2018)
gRPC network management interface (gNMI).
Internet-Draft
Technical Report draft-openconfig-rtgwg-gnmi-spec-01, IETF.
External Links: Link (https://datatracker.ietf.org/doc/html/draft-openconfig-rtgwg-gnmi-spec-01)
Cited by: §5.2, §6.1, §7.3, §7.7.

[61] J. Shalf (2020)
The future of computing beyond Moore’s law.
Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences **378** (2166), pp. 20190061.
External Links: Document (https://dx.doi.org/10.1098/rsta.2019.0061), Link (https://royalsocietypublishing.org/doi/10.1098/rsta.2019.0061), ISSN 1364-503X
Cited by: §4.1.

[62] A. Shehabi, S. J. Smith, A. Hubbard, A. Newkirk, N. Lei, M. A. B. Siddik, B. Holecek, J. Koomey, E. Masanet, and D. Sartor (2024)

2024 United States Data Center Energy Usage Report.
Technical report
Technical Report LBNL-2001637, Lawrence Berkeley National Laboratory, Berkeley, California.
Note: Accessed: 2026-01-23
External Links: Link (https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf)
Cited by: §1.2, §1, §2.1, §2.2, §2.2, §3.1, §3.1, §3.2, §5.1.

[63] S. Shekhar, W. Bogaerts, L. Chrostowski, J. E. Bowers, M. Hochberg, R. Soref, and B. J. Shastri (2024)
Roadmapping the next generation of silicon photonics.
Nature Communications **15** (1), pp. 751.
External Links: Document (https://dx.doi.org/10.1038/s41467-024-44750-0), Link (https://doi.org/10.1038/s41467-024-44750-0)
Cited by: §7.2.

[64] C.A. Silva, R. Vilaça, A. Pereira, and R.J. Bessa (2024)
A review on the decarbonization of high-performance computing centers.
Renewable and Sustainable Energy Reviews **189**, pp. 114019.
External Links: ISSN 1364-0321, Document (https://dx.doi.org/https%3A//doi.org/10.1016/j.rser.2023.114019), Link (https://www.sciencedirect.com/science/article/pii/S1364032123008778)
Cited by: §8.1, §8.1, §8.4.

[65] A. Singla, B. Chandrasekaran, P. B. Godfrey, and B. Maggs (2014)
The Internet at the speed of light.
In Proceedings of the 13th ACM Workshop on Hot Topics in Networks (HotNets-XIII), Los Angeles, CA, pp. 1–7.
External Links: Document (https://dx.doi.org/10.1145/2670518.2673876)
Cited by: §4.1, §4.

[66] A. Tandon, A. Dhir, A.K.M. N. Islam, and M. Mäntymäki (2020)
Blockchain in healthcare: a systematic literature review, synthesizing framework and future research agenda.
Computers in Industry **122**, pp. 103290.
External Links: ISSN 0166-3615, Document (https://dx.doi.org/https%3A//doi.org/10.1016/j.compind.2020.103290), Link (https://www.sciencedirect.com/science/article/pii/S0166361520305248)
Cited by: §7.5, §7.5.

[67] Telecom Infra Project (2024)
GNPy: Optical Route Planning and Quality-of-Transmission Estimation.
Note: Open-source library, Telecom Infra Project. Available at: https://gnpy.readthedocs.io (https://gnpy.readthedocs.io)
External Links: Link (https://gnpy.readthedocs.io)
Cited by: §8.1.

[68] The Green Grid (2007)
Green grid metrics: describing data center power efficiency.
Technical report
The Green Grid.
External Links: Link (https://www.thegreengrid.org)
Cited by: §1.1.

[69] J. J. Tithi, H. Wu, A. Abuhatzera, and F. Petrini (2025)
Scaling intelligence: designing data centers for next-gen language models.
External Links: 2506.15006, Document (https://dx.doi.org/10.48550/arXiv.2506.15006), Link (https://arxiv.org/abs/2506.15006)
Cited by: §4.1, §4.5.

[70] TM Forum (2024)
Autonomous networks technical architecture.
TM Forum.
Note: https://www.tmforum.org/resources/toolkit/autonomous-networks-technical-architecture/ (https://www.tmforum.org/resources/toolkit/autonomous-networks-technical-architecture/)Technical architecture toolkit, IG1230
Cited by: §5.2, §6.1, §6.1, §7.4.

[71] B. Toni and T. Jones (2025)
Sovereignty, security, scale: a UK strategy for AI infrastructure.
Tony Blair Institute for Global Change.
Note: https://institute.global/insights/tech-and-digitalisation/sovereignty-security-scale-a-uk-strategy-for-ai-infrastructure (https://institute.global/insights/tech-and-digitalisation/sovereignty-security-scale-a-uk-strategy-for-ai-infrastructure)Policy report
Cited by: §2.4.

[72] United States Congress (2018)
Clarifying lawful overseas use of data act (CLOUD act), H.R. 4943, 115th Congress.
Note: Pub. L. No. 115-141, div. V, 132 Stat. 348
External Links: Link (https://www.congress.gov/bill/115th-congress/house-bill/4943)
Cited by: §1.1.

[73] W3C (2023)
Trace context.
World Wide Web Consortium.
Note: https://www.w3.org/TR/trace-context/ (https://www.w3.org/TR/trace-context/)Accessed: 2026-03-30
Cited by: §7.5.

[74] T. Xiao and F. You (2025)
Environmental impact and net-zero pathways for sustainable artificial intelligence servers in the USA.
Nature Sustainability.
External Links: Document (https://dx.doi.org/10.1038/s41893-025-01681-y), Link (https://www.nature.com/articles/s41893-025-01681-y)
Cited by: §1.2, §3.1, §3.3, §3.4, §3.5, Figure 7, §8.1, §8.1, §8.2, §8.5.

[75] M. Yñez-Barnuevo and Environmental and Energy Study Institute (EESI) (2025)
Data centers and water consumption.
Note: Accessed: 2026-01-30
External Links: Link (https://www.eesi.org/articles/view/data-centers-and-water-consumption)
Cited by: §3.4, Figure 7, §8.1, §8.1, §8.1.

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